

Indian Knowledge System



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Unit 1

Defining Indian Knowledge System. Importance of Ancient Knowledge, The IKS Corpus. Introduction to Traditional Knowledge: Concept & Definition, Nature, Scope and Importance of Traditional Knowledge. The Vedic Corpus: Introduction to Vedas, The Four Vedas. Philosophical System: Development and Unique Features. Bhagavad Gita

Approximately 3,000 years of contributions from scholars, artisans, physicians, and philosophers have established the Indian subcontinent as a cradle of knowledge that has evolved continuously over time. The Indian Knowledge System is being considered not as a collection of isolated beliefs, but as a strong value-based system built on observation, reasoning, and practical experience. Its approaches to science, philosophy, society, and nature were deeply interconnected. As present-day challenges demand multidisciplinary approaches, these traditions offer a holistic lens, positioning the Indian Knowledge System as an academic field with growing value across education, research, and present-day practice.

The following is why the study of Indian Knowledge Systems is opportune in the modern world:

- ✓ It offers a holistic approach to learning that integrates ethics, knowledge, and real-world implementation.
- ✓ It helps beginners comprehend the history of the Indian knowledge system beyond textbook stories.
- ✓ It establishes the clarity of basics by the basic structure of the Indian knowledge system, functional in cross-disciplinary.
- ✓ It offers alternative viewpoints that merge with modern science and critical thinking.
- ✓ It pre-language and prepares learners to participate globally in discussions about sustainability and cultural knowledge.





Defining Indian Knowledge System:

The Indian Knowledge System (IKS) refers to the vast and diverse body of knowledge that has been developed and transmitted in India over millennia. It encompasses a wide range of disciplines, including philosophy, mathematics, astronomy, medicine, arts, and spiritual practices.

Cāṇakya, in his Arthaśāstra, classifies knowledge that one needs to know into four types:

- ✓ ānvīkṣikī – used in the general sense of (contemporary) Science, encompassing logic and philosophy
- ✓ trayī – used to refer to traditional knowledge, to the Vedas in particular
- ✓ vārtā – used to represent wealth creation (agriculture and commerce)
- ✓ daṇḍanīti – used to represent politics and public administration

INDIAN civilization has always attached great value to knowledge. In Srimad Bhagavad Gita, 4.33, 37-38, Lord Krishna tells Arjuna that knowledge is the great purifier and liberator of the self. India's knowledge tradition is ancient and uninterrupted like the flow of the river Ganga, from the Vedas (Upanisads) to Sri Aurobindo, knowledge or Jnana has been at the centre of all rational and speculative inquiry in India.

Three terms are closely connected in all discussions of knowledge — Darsana, Jnana and Vidya. Darsana, philosophy is the “system”, the point of view, which yields/leads to Jnana, knowledge.

When knowledge gathered about a particular domain is organized and systematized for purposes of, say, reflection and pedagogy, it is called Vidya, “discipline”.

Key Features of the Indian Knowledge System:

1. Interdisciplinary Nature: IKS integrates various fields of knowledge, showcasing a holistic approach to learning and understanding the universe.

2. Ancient Texts and Oral Traditions: Much of IKS is rooted in ancient texts such as the Vedas, Upanishads, and classical treatises in various sciences, as well as in oral traditions passed down through generations.

3. Cultural Context: IKS is deeply intertwined with the cultural, social, and historical contexts of India, reflecting its rich heritage and diverse practices.

4. Philosophical Foundations: Indian philosophy, with its emphasis on concepts such as dharma (duty), karma (action), and moksha (liberation), plays a crucial role in shaping IKS.

5. Contemporary Relevance: Today, IKS is gaining recognition for its potential contributions to modern science, environmental sustainability, and holistic health practices.

Importance of Ancient Knowledge:

The importance of ancient knowledge lies in its foundational role in shaping contemporary thought, culture, and practices. Here are several key points highlighting its significance:

1. Cultural Heritage: Ancient knowledge encapsulates the wisdom and experiences of past civilizations, providing insights into their beliefs, customs, and social structures. It helps preserve cultural identity and fosters a sense of belonging.

2. Scientific Foundations: Many modern scientific principles and methods have roots in ancient practices. For example, early advancements in mathematics, astronomy, and medicine laid the groundwork for contemporary disciplines.

3. Philosophical Insights: Ancient philosophies offer profound perspectives on ethics, existence, and the nature of knowledge. These ideas continue to influence modern philosophy, psychology, and spirituality.

4. Sustainable Practices: Ancient knowledge often emphasizes harmony with nature and sustainable living. Traditional ecological practices can inform modern approaches to environmental conservation and resource management.

5. Innovation and Adaptation: By studying ancient knowledge, contemporary thinkers and innovators can draw inspiration for new solutions to current problems, fostering creativity and interdisciplinary collaboration.

6. Holistic Understanding: Ancient systems of knowledge often promote a holistic view of life, integrating mind, body, and spirit. This approach can enhance well-being and inform practices in health and education.

The IKS Corpus:

The Indian Knowledge System (IKS) Corpus refers to the comprehensive collection of knowledge, texts, practices, and philosophies that have evolved in India over centuries. It includes a wide range of disciplines and cultural expressions, reflecting the depth and diversity of Indian thought. Here are the key components of the IKS Corpus:

1. Ancient Texts: This includes foundational texts such as the Vedas, Upanishads, Puranas, and Dharmashastras, which encompass various fields like philosophy, spirituality, law, and ethics.

2. Mathematics and Astronomy: The contributions of ancient Indian scholars, such as Aryabhata and Brahmagupta, in mathematics and astronomy are significant. Their work laid the groundwork for concepts such as zero and algebra.

3. Ayurveda and Traditional Medicine: Ancient Indian medical systems, particularly Ayurveda, provide holistic approaches to health, emphasizing balance and prevention, and are increasingly recognized for their relevance in modern healthcare.

4. Arts and Aesthetics: The IKS Corpus includes classical art forms, literature, music, and dance. These art forms reflect cultural narratives and spiritual expressions, enriching the understanding of Indian aesthetics.

5. Philosophical Traditions: Various schools of thought, such as Vedanta, Buddhism, Jainism, and Nyaya, contribute to the philosophical richness of the corpus, exploring questions of existence, consciousness, and morality.

6. Oral Traditions: Many knowledge systems were transmitted orally, including folklore, stories, and teachings, which play a crucial role in preserving cultural heritage and values.

7. Sustainable Practices: Traditional ecological knowledge, including farming techniques, water management, and resource conservation, is integral to the IKS, emphasizing sustainable living in harmony with nature.

Introduction to Traditional Knowledge:

Concept & Definition:

Traditional Knowledge (TK) refers to the collective wisdom and practices developed by communities over generations, often rooted in their cultural, historical, and environmental contexts. In the Indian Knowledge System (IKS), TK encompasses a wide array of knowledge forms, including spiritual, agricultural, medicinal, and artistic traditions, reflecting the rich tapestry of India's diverse cultures.

The most important means of knowledge, according to *Nyāya darśana*, called *pramāṇa* (literally evidence, or mā = measure) are as follows:

1. Pratyakṣa (direct perception through the five sensory inputs: sound, touch, sight, taste and smell)
2. Anumāna (inference)
3. Upamāna (example)
4. Arthāpatti (2-stepped inference)
5. Anupalabdhi (non-perception)
6. Śabda (Verbal testimony)

Different darśanas accept various pramāṇas listed above to establish their point of view.

Nature of Traditional Knowledge:

- Culturally Embedded: In India, TK is closely tied to specific communities and their unique identities, incorporating local languages, rituals, and customs.
- Dynamic and Evolving: While TK often appears static, it is continually adapted and transformed in response to changing social, economic, and environmental conditions.
- Holistic and Interconnected: Indian TK emphasizes the interconnectedness of various domains, integrating spirituality, ecology, health, and social practices into a cohesive worldview.

Scope of Traditional Knowledge:

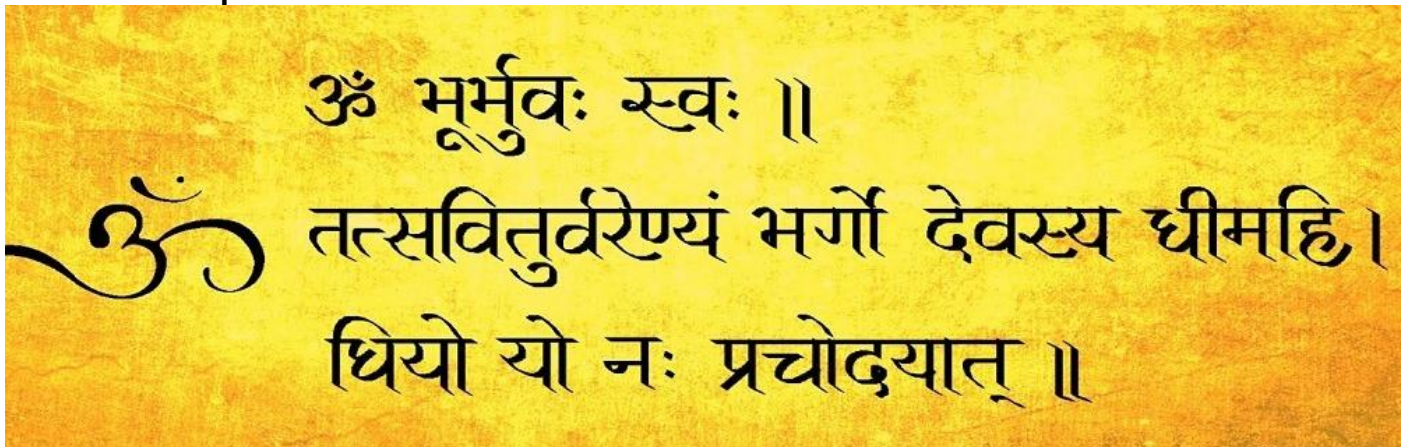
The scope of TK within the IKS is extensive, encompassing various domains, including:

1. Agriculture: Traditional farming practices, seed preservation techniques, and sustainable land management methods developed over centuries.
2. Ayurveda and Indigenous Medicine: Comprehensive systems of health that utilize herbal remedies, dietary practices, and holistic healing approaches.
3. Philosophical Traditions: Deep philosophical insights from schools like Vedanta, Buddhism, and Jainism, addressing ethics, existence, and the nature of knowledge.
4. Arts and Crafts: Rich traditions of music, dance, visual arts, and craftsmanship that convey cultural narratives and communal values.
5. Ecological Knowledge: Indigenous practices related to resource management, conservation of biodiversity, and sustainable use of natural resources.

Importance of Traditional Knowledge:

1. Cultural Heritage Preservation: TK serves as a repository of cultural identity, ensuring that languages, customs, and practices are preserved for future generations.
2. Sustainable Development: Traditional ecological knowledge provides insights into sustainable practices that promote environmental conservation and biodiversity, which are increasingly relevant in the face of climate change.
3. Holistic Health Approaches: The integration of TK in health practices, such as Ayurveda, offers alternative and complementary healing modalities that emphasize prevention and well-being.
4. Social Resilience: Communities that leverage traditional knowledge often exhibit greater resilience, drawing on their cultural practices to navigate social and environmental challenges.
5. Innovation and Knowledge Sharing: TK can inspire new ideas and solutions by blending ancient wisdom with modern science, facilitating innovative approaches to contemporary issues.
6. Recognition and Rights: Acknowledging and protecting traditional knowledge is vital for ensuring that local communities benefit from their cultural heritage and innovations, safeguarding their rights against exploitation.

The Vedic Corpus:



The Vedic corpus represents a profound heritage of spiritual and cultural knowledge. Together, they not only establish the foundational concepts of Hindu philosophy and practice but also offer a window into the social, cosmological, and ethical frameworks of ancient Indian civilization. The Vedas continue to be a source of inspiration, study, and reverence in contemporary spiritual and philosophical discourse.

Introduction to Vedas:

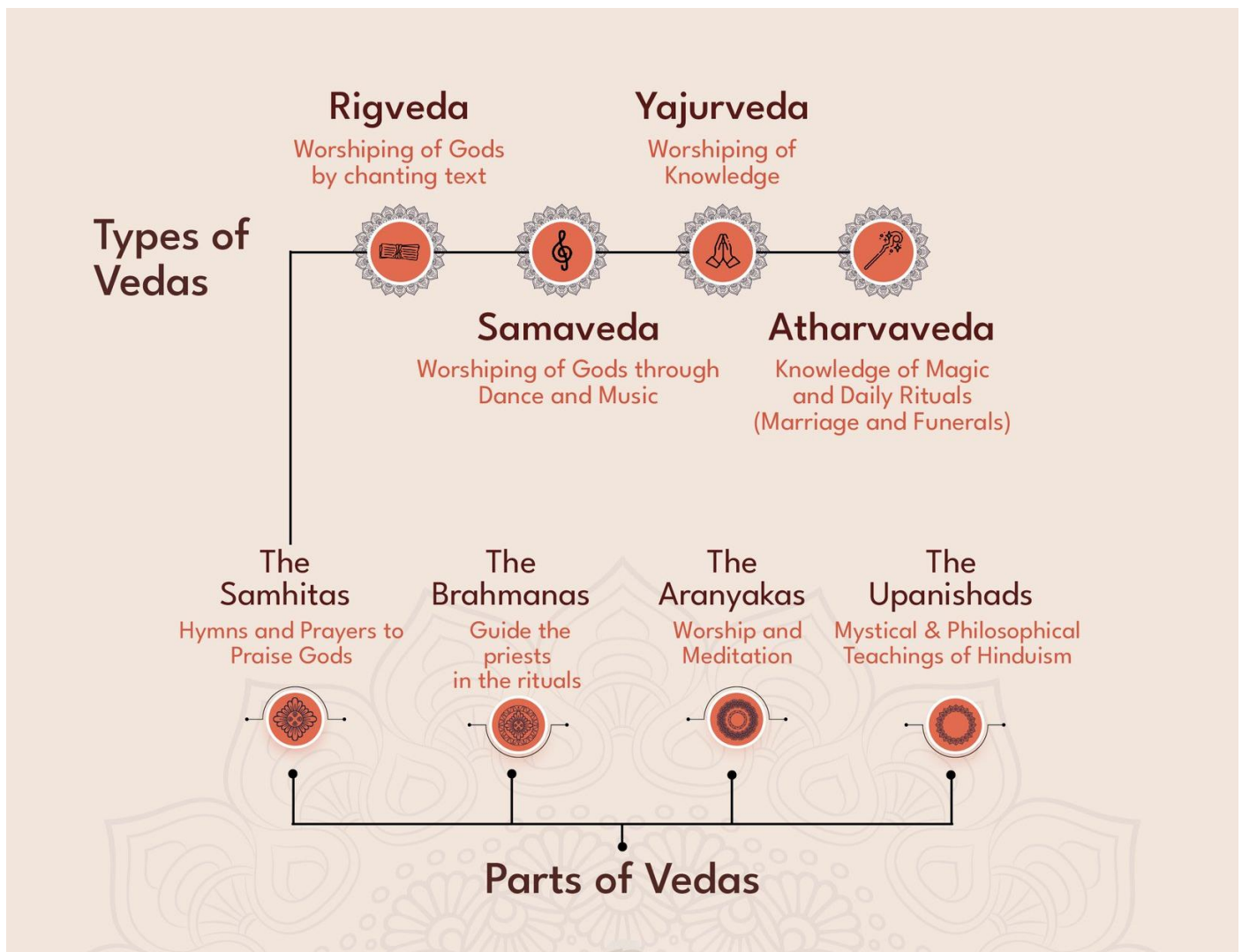
The Vedas are a collection of ancient texts that form the foundational scriptures of Hinduism and represent some of the oldest known writings in the world. Composed in Sanskrit, these texts are revered as sacred knowledge (*śruti*), believed to be divinely revealed to ancient sages (*rishis*). The *śruti* represents timeless and eternal values which hold good for all times, while the *smṛti* codifies these eternal values into codes or rules based on the needs of the time and place for the society at hand. The Vedas encompass a wide range of topics, including rituals, philosophy, hymns, and cosmology, reflecting the spiritual, cultural, and social dimensions of early Indian civilization. The *Vedas* are said to be *apauruṣeya*; or in other words, the Vedas are *vastu-niṣṭha* (in tune with reality / existence) and not *vyakti-niṣṭha* (one's emotions, likes and dislikes, orientations).

The Vedas consist of four major books – *R̥gveda*, *Yajurveda*, *Sāmaveda* and *Atharvaṇaveda*. Each Veda is further divided into four parts – Samhita (hymns), Brahmanas (rituals), Aranyakas (meditation), and Upanishads (philosophical discussions). The Upanishads, also known as the Vedānta, are considered to be the final part of the Vedas and are the basis of Hindu philosophy. The Upanishads contain profound discussions on the nature of reality, the purpose of life, and the relationship between the individual soul (*Ātman*) and the universal consciousness (*Brahman*).

Significance of the Vedas:

The Vedas are not just religious texts; they are also critical historical documents that provide insights into the beliefs, practices, and societal structures of ancient India. They are considered the source of many Hindu doctrines and rituals and have influenced various philosophical systems and cultural practices throughout Indian history.

Since the *Vedas* are a repository of knowledge in such varied disciplines; one must take support of the six *Vedāṅgas* in order to make an attempt to decipher the meaning of the Vedic *mantras*.



The Four Vedas:

1. Rigveda:

- **Overview:** The Rigveda is the oldest and most important of the four Vedas, consisting of 1,028 hymns (suktas) dedicated to various deities. It is primarily a collection of prayers and invocations.
- **Content:** The hymns are addressed to natural forces and deities like Agni (fire), Indra (rain and thunder), and Varuna (cosmic order). It also contains philosophical inquiries about existence and the universe.
- **Significance:** The Rigveda lays the groundwork for Vedic rituals and cosmology and is a key text for understanding early Indo-Aryan culture.

2. Samaveda:

- **Overview:** The Samaveda consists mainly of melodies and chants derived from the Rigveda, specifically designed for musical performance during rituals.
- **Content:** It contains verses (saman) arranged for chanting and is essential for the performance of sacrifices (yajnas) where music plays a crucial role.
- **Significance:** The Samaveda emphasizes the importance of sound and music in Vedic rituals, influencing Indian classical music traditions.

3. Yajurveda:

- **Overview:** The Yajurveda is a compilation of ritual formulas (yajus) used in sacrifices and contains both prose and poetry. It serves as a guide for performing rituals.
- **Content:** It includes instructions on how to conduct rituals, the materials needed, and the mantras to be recited. There are two main texts (divisions/versions):

1. **Shukla Yajurveda** (White Yajurveda) — also called Vajasaneyi Samhita. It keeps the mantras separate from the explanatory Brahmana material.
 2. **Krishna Yajurveda** (Black Yajurveda) — its Samhita mixes the mantras with explanatory Brahmana prose.
- Significance: The Yajurveda bridges the gap between the theoretical knowledge of the Rigveda and the practical application of rituals, making it essential for priests and practitioners.

4. Atharvaveda:

- Overview: The Atharvaveda is distinct from the other three Vedas, containing hymns and spells that address everyday life, health, and societal issues.
- Content: It includes magical incantations, medicinal formulas, and philosophical hymns, reflecting a broader understanding of life beyond ritualistic practices.
- Significance: The Atharvaveda highlights the practical aspects of Vedic knowledge and the importance of addressing the concerns of common people, making it unique among the Vedas.

Philosophical System:

The philosophical systems within the Indian Knowledge System represent a rich and diverse tapestry of thought that has evolved over millennia. With their holistic approach, emphasis on experience, and ethical foundations, these philosophies continue to provide valuable insights into the nature of existence, knowledge, and the human experience. They encourage a dialogue between tradition and modernity, offering timeless wisdom relevant to contemporary challenges.

Although many people translate *darśana* as 'philosophy', it is not really satisfying as philosophy is etymologically derived as *philos* (φίλος 'beloved/loving') *sophia* (σοφία 'wisdom'), essentially meaning 'love for wisdom'; whereas *darśana* is much more. Literally, *darśana* means 'vision' and may be called as a 'system of thought' which propounds a vision of the truth of existence.

Development of Indian Philosophical Systems:

There are six major (āstika or orthodox) schools of Vedic philosophy—Nyaya, Vaisheshika, Samkhya, Yoga, Mīmāṃsā and Vedānta — and five major non-Vedic or heterodox (nāstika or sramanic) schools—Jain, Buddhist, Ajivika, Ajñāna, and Charvaka. The āstika group embraces the Vedas as an essential source of its foundations, while the nāstika group does not. However, there are other methods of classification; Vidyananda (Mādhavācārya) for instance identifies sixteen schools of Indian philosophy by including those that belong to the Śaiva and Raseśvara traditions.

Indian philosophical systems have evolved over thousands of years, deeply rooted in the spiritual and cultural fabric of the subcontinent. The development of these philosophies can be traced through several key phases:

1. Vedic Period: The earliest philosophical ideas emerged in the Vedas, particularly the Upanishads, which introduced concepts such as Brahman (the ultimate reality) and Atman (the self). This period laid the groundwork for later philosophical inquiry.

2. Classical Period: Following the Vedic texts, various schools of thought began to develop, including the six orthodox (āstika) schools of philosophy: Nyaya (logic), Vaisheshika (atomism), Samkhya (enumeration), Yoga (union), Mimamsa (ritualistic exegesis), and Vedānta (end of the Vedas). Each of these systems explored fundamental questions about existence, knowledge, and ethics.

3. Buddhism and Jainism: In the 6th century BCE, new philosophical traditions arose with the teachings of the Buddha and Mahavira, emphasizing concepts like impermanence, non-

attachment, and ahimsa (non-violence). These traditions contributed significantly to Indian philosophical discourse, often challenging and expanding upon Vedic thought.

4. Medieval to Modern Period: The interaction between Indian philosophies and Islamic and Western thought during the medieval period led to the synthesis of ideas and the emergence of new philosophical inquiries. Modern Indian philosophers began to reinterpret traditional concepts in light of contemporary challenges.

Unique Features of Indian Philosophical Systems:

1. Holistic Approach: Indian philosophy often emphasizes a holistic understanding of existence, integrating metaphysics, ethics, epistemology, and spirituality. This contrasts with some Western traditions that may compartmentalize these domains.

2. Unity of Opposites: Many Indian philosophical systems explore the unity of opposites, such as the interplay between the individual self (Atman) and the universal spirit (Brahman) in Vedanta. This reflects a nuanced understanding of duality and interconnectedness.

3. Focus on Experience: Indian philosophy places significant importance on personal experience and realization. Practices like meditation and yoga are integral to many philosophical traditions, emphasizing experiential knowledge over purely theoretical understanding.

4. Diversity of Thought: The Indian philosophical landscape is marked by a remarkable diversity of perspectives. Various schools offer different paths to understanding reality, such as the dualism of Samkhya and the non-dualism of Advaita Vedanta, demonstrating an openness to multiple viewpoints.

5. Emphasis on Ethical Living: Ethical considerations are central to many Indian philosophical systems. Concepts such as Dharma (duty/righteousness) and Ahimsa (non-violence) guide moral conduct and social responsibility, reflecting a commitment to harmonious living.

6. Interrelation of Knowledge Systems: Indian philosophy often intersects with other fields of knowledge, including science, art, and spirituality. This interconnectedness fosters a comprehensive worldview, recognizing that different domains of knowledge contribute to a greater understanding of life.

Schools of Indian Philosophy:

Indian Philosophy encompasses a wide range of thought systems that explore metaphysics, epistemology, ethics, logic, and the nature of reality. These philosophies were not merely speculative but also deeply practical, guiding how one should live, act, and seek liberation (moksha).

The traditional classification of Indian philosophy divides it into two broad categories:

1. Āstika (Orthodox) Schools – accept the authority of the Vedas
2. Nāstika (Heterodox) Schools – reject the authority of the Vedas

The Āstika (Orthodox) Schools:

These schools recognize the Vedas as the ultimate source of knowledge and authority. Each school has its own philosophical system but often complements others in some areas.

- Nyāya (Logic and Epistemology)
 - Focus: Logic, critical thinking, and methods of knowledge acquisition.
 - Founder: Sage (Rishi) Gautama
 - Contribution: Developed tools of logical analysis (syllogism, inference).
 - Goal: Attain liberation through right knowledge.
- Vaiśeṣika (Atomism and Metaphysics)
 - Focus: Categorization of reality; metaphysics.
 - Founder: Sage Kaṇāda
 - Contribution: Proposed that all objects are made of atoms (anu).

- Six categories (Padārthas): Substance, quality, activity, generality, particularity, and inherence.
- Sāṅkhya (Dualism)
 - Focus: Enumeration of reality; dualistic philosophy.
 - Founder: Sage Kapila
 - Core Belief: Two eternal realities—Purusha (consciousness) and Prakriti (matter).
 - Goal: Liberation through discriminative knowledge of Purusha and Prakriti.
- Yoga (Discipline and Practice)
 - Focus: Practical path to liberation through physical and mental discipline.
 - Founder: Sage Patañjali (Yoga Sūtras)
 - Related to: Sāṅkhya school, but adds Ishvara (God) as a central figure.
 - Eightfold Path (Ashtanga Yoga): Includes ethical rules, postures, breath control, meditation, and more.
- Mīmāṃsā (Ritualism and Dharma)
 - Focus: Interpretation of the Vedas, especially rituals.
 - Founder: Sage Jaimini
 - Goal: Liberation through the performance of dharma (righteous actions).
 - Core Idea: The Vedas are eternal and self-evident.
- Vedānta (Uttara Mīmāṃsā – the “End of the Vedas”)
 - Focus: Interpretation of Upanishads, Brahma Sūtras, and Bhagavad Gītā (the *Prasthānatrayī*).
 - Founder: Sage Bādarāyaṇa (author/compiler of the Brahma Sūtras).
 - Goal: Liberation through realizing Ātman–Brahman unity or relationship, depending on the school.
 - Core Idea: Ultimate reality is Brahman; Self (Ātman) is fundamentally connected to or identical with Brahman; the phenomenal world may be illusory (Māyā) — seeking mokṣa from the cycle of rebirth.

The Nāstika (Heterodox) Schools:

These philosophies reject the authority of the Vedas, yet they present complete systems of thought, including ethics, metaphysics, and paths to liberation.

- Jainism
 - Founder: Mahāvīra (24th Tirthankara)
 - Core Beliefs: Non-violence (Ahimsā), Multiplicity of viewpoints (Anekāntavāda), Karma as a material substance
 - Goal: Liberation through self-restraint, asceticism, and non-violence.
 - Rejects: Vedas, caste hierarchy, creator God.
 - Famous quotes/rituals: Kṣamāvaṇī (Digambaras) or Samvatsari (Śvētāmbaras), The phrase "Micchami Dukkadam".
- Buddhism
 - Founder: Siddhārtha Gautama (The Buddha)
 - Core Beliefs: Four Noble Truths (suffering, cause, cessation, path), Eightfold Path, No-Self (Anattā) and Impermanence (Anicca)
 - Goal: Liberation (Nirvāṇa) through ethical living and meditation.
 - Rejects: Vedic authority, caste system, eternal self (Ātman).
 - Famous quotes/rituals:
 - The *Tripitaka* (Vinaya Pitaka, Sutta Pitaka, Abhidhamma Pitaka)
 - The *pañcaśīla* (five precepts): Abstain from killing, Abstain from theft, Abstain from sexual misconduct, Abstain from falsehood, Abstain from intoxication
 - The Eightfold Path: right view, right intention, right speech, right action, right livelihood, right effort, right mindfulness, and right concentration
- Cārvāka (Materialism)
 - Focus: Only the material world exists; direct perception is the only valid knowledge.
 - Belief: Rejection of the soul, afterlife, karma, and liberation.
 - Slogan: “Eat, drink, and be merry!” – A purely hedonistic view of life.

- Contribution: Served as a rational critique of religious orthodoxy.
- Famous quotes/rituals:
यावत् जीवेत् सुखम् जीवेत्। ऋणं कृत्वा घृतं पिबेत्। भस्मिभूतस्य देहस्य पुनरागमनं कुतः॥
पीत्वा पीत्वा पुनः पीत्वा, यावत् पतति भूतले। उत्थाय च पुनः पीत्वा, पुनर्जन्म न विद्यते॥

➤ Ājīvika

- Founder: Makkhali Gosāla (contemporary of Mahāvīra and the Buddha)
- Core Beliefs: Absolute Determinism (Niyati), Denial of Karma's Efficacy, eternal soul
- Slogan: "Everything is preordained".
- Criticism: "Fatalism is the lazy man's way of accepting the inevitable." – Natalie Clifford Barney

➤ Ajñāna

- Ajñāna means "non-knowledge" or "agnosticism".
- Mentioned in Buddhist texts such as the Sāmaññaphala Sutta, Brahmajala Sutta and Sūyagaḍaṃga of Jainism.
- Associated with Śramaṇa wanderers.
- Core Beliefs: True knowledge is impossible, no ethical or spiritual claim is absolute.
- Rejects: existence, soul, afterlife, or liberation

The Bhagavad Gita in the Context of Indian Knowledge System:



The Bhagavad Gita, often referred to simply as the Gita, is a 700-verse scripture that is part of the Indian epic Mahabharata. It is a philosophical and spiritual dialogue between Prince Arjuna and Lord Krishna, who serves as his charioteer. The Gita addresses profound questions about duty, righteousness, and the nature of reality, making it a central text in the Indian Knowledge System.

Key Themes of the Bhagavad Gita:

1. Dharma (Righteous Duty): Central to the Gita is the concept of dharma, which refers to the moral and ethical duties specific to an individual's role in society. Arjuna's initial reluctance to fight in the Kurukshetra War represents a conflict between personal feelings and his Kshatriya duty. Krishna guides him to understand that fulfilling one's dharma, even in the face of adversity, is essential for the maintenance of cosmic order.

2. Yoga (Paths to Realization): The Gita outlines various paths of yoga, including:

- Karma Yoga: The path of selfless action, emphasizing that one should act without attachment to the results.
- Bhakti Yoga: The path of devotion, highlighting the importance of surrendering to God and cultivating love and devotion.
- Jnana Yoga: The path of knowledge, focusing on the pursuit of wisdom and understanding the true nature of reality.

3. Self and Reality: The Gita explores the distinction between the eternal self (Atman) and the physical body. It teaches that the Atman is immortal and transcends physical existence, thus encouraging individuals to rise above material concerns.

4. Equanimity (Dhairya): The Gita emphasizes maintaining equanimity in both success and failure, pleasure and pain. This balanced mindset is essential for spiritual growth and fulfillment of one's duties.

5. Divine Guidance: Krishna's role as a divine guide illustrates the importance of seeking wisdom and guidance from higher consciousness. The relationship between Arjuna and Krishna exemplifies the teacher-student dynamic that is vital in Indian philosophical and spiritual traditions.

Importance of the Bhagavad Gita in the Indian Knowledge System:

1. Philosophical Foundation: The Gita synthesizes various philosophical concepts from the Vedic texts, including ideas from Samkhya and Vedanta, and presents them in a practical context. This synthesis has made it a critical resource for understanding Indian philosophy.

2. Ethical and Moral Framework: The Gita's teachings on dharma and ethics provide a framework for moral decision-making, making it relevant for individuals navigating personal and societal challenges.

3. Spiritual Guidance: The Gita serves as a spiritual guide for practitioners of various paths, offering insights into meditation, devotion, and the nature of the self. Its teachings are applicable to both spiritual seekers and those engaged in worldly life.

4. Cultural Impact: The Gita has significantly influenced Indian culture, art, literature, and political thought. It has inspired countless leaders, thinkers, and movements, including figures like Mahatma Gandhi, who drew upon its principles in the struggle for independence.

5. Interdisciplinary Relevance: The concepts in the Gita extend beyond religious boundaries, influencing fields such as psychology, leadership, and conflict resolution. Its teachings on mindfulness, action without attachment, and inner peace resonate with contemporary issues.

The Bhagavad Gita is a seminal text within the Indian Knowledge System, offering profound insights into ethics, duty, and spirituality. Its teachings remain relevant across time and culture, serving as a guide for personal development, ethical living, and understanding the complexities of existence. Through its philosophical depth and practical wisdom, the Gita continues to inspire individuals on their spiritual journeys and provides a foundation for exploring the interconnectedness of life.

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Unit 2

The Vedangas and Other Streams of Indian Knowledge System: The six Vedangas — Siksha, Vyakarana, Chandas, Nirukta, Jyotisha and Kalpa.

The Vedangas and Other Streams of the Indian Knowledge System:

The Indian Knowledge System (IKS) is a vast and intricate framework encompassing various disciplines, philosophies, and practices that have evolved over millennia. Among these, the Vedangas and other streams of knowledge play a crucial role in shaping the intellectual landscape of India.

The Vedangas:

The Vedangas are the last treatises of the Vedic Literature. Paniniya Shiksha (41-42) narrates two verses on the importance of the Vedangas which describe Veda as a Purusha having six limbs as six Vedangas: Chandas are His two feet, Kalpa are His two arms, Jyotisha are His eyes, Nirukta is His ears, Shiksha is His nose and Vyakarana is His mouth. The oldest record of their names occurs in the Mundaka Upanishad (1.1.5) where they are named as:

1. Śikṣā or phonetics or pronunciation
2. Kalpa or ritual
3. Vyākaraṇa or grammar
4. Nirukta or etymology
5. Chandas or meter
6. Jyotiṣa or astronomy

1. Shiksha (Phonetics or pronunciations): This discipline focuses on the correct pronunciation, intonation, and articulation of Vedic texts. It emphasizes the sound aspects of language, ensuring that mantras are chanted accurately to preserve their efficacy.

Shiksha really means instruction: then in particular 'instruction in reciting' i.e., in correct pronunciation, accentuation etc. of the Samhita texts. Later, it was a name given to works containing rules regarding the proper pronunciation of Vedic texts. Thus, the Shiksha- Sutras are treatises on phonetics. They are related to the Samhita and, therefore, are almost as old as the Kalpa-Sutras.

Shiksha lays down the rules of phonetics - sounds of syllables, of pronunciation. The function of the Shiksha is thus to fix the parameters of Vedic words. Phonetics is most important in the case of the Vedic language, because we see that change in sound leads to change in results and effect. Hence, Shiksha which is Vedic Phonetics has been regarded as the most important of the six Angas (organs) of the Veda Purusha. Some important *Pratishakhyas* are:

- (1) Rigveda - Pratishakhya of Rigved
- (2) Taittiriya - Pratishakhya of Krishna Yajurveda
- (3) Vajasaneyi - Pratishakhya of Shukla Yajurveda
- (4) Atharvaveda - Pratishakhya of Atharvaveda

2. Kalpa (Rituals): Kalpa encompasses the rules and procedures for performing Vedic rituals and sacrifices. It includes detailed guidelines on various ceremonies, rites, and the ethical conduct of priests, ensuring the correct performance of religious practices.

The second Vedanga, Kalpa is called the arms of the Veda Purusha. It is especially intended for the proper application of the Vedic texts. The oldest Kalpa-Sutras are those which in their contents are directly connected with the Brahmanas and Aranyakas. It was the ritual (Kalpa), the chief

contents of the Brahmanas, which first received systematic treatment in the manuals called the Kalpa-Sutras. They contain the rules in the Sutra style, referring to sacrifices, with the omission of all things which are not immediately connected with the ceremonial. They are more practical than the Brahmanas which for the most part are taken up with mystical, historical, mythological, etymological and theological discussions. They are also considered significant for the study of Vedic culture and society.

There are four types of the Kalpa-Sutras:

- (1) Shrauta-Sutras, dealing with Shrauta sacrifices
- (2) Grihya-Sutras, dealing with the domestic ceremonies
- (3) Dharma-Sutras, dealing with the religious and social laws
- (4) Shulba-Sutras, dealing with the rules of measurement of the fire-altars etc.

3. Vyakarana (Grammar): This Vedanga studies the grammar and syntax of the Sanskrit language. It provides rules for constructing sentences, understanding meanings, and analysing texts, thereby facilitating precise communication and interpretation.

Vyakarana is called the mouth of the Veda Purusha. The old Vedanga-texts on Vyakarana are entirely lost today. In the Aranyakas, we find some technical terms of grammar. The only representative of this Vedanga is the Ashtadhyayi of Panini, which belongs to a later period. It is indeed the most celebrated text-book of grammar. It is not associated with any Vedic school. Due to its great merits, this may be assumed that Panini superseded all his predecessors, whose works have consequently perished.

Formation of the word is the main subject of grammar. It discusses root (Prakriti) and suffix (Pratyaya) of a word to study its meaning. Panini's Vyakarana is in the form of sutras or aphorisms. The fourteen Sutras are referred to here, as Maheswara Sutras. They were originated from Nataraja's damuru sound. They are considered the foundation of grammar. Vararuci has written an elaborate commentary or Vartika. Sage Patanjali wrote commentary or Bhashya on it.

4. Nirukta (Etymology): Nirukta deals with the interpretation of Vedic words and phrases through etymology. It aims to elucidate the meanings of terms used in the Vedas, enhancing the understanding of their philosophical and spiritual significance.

Nirukta Vedanga is called the ears of the Veda Purusha. 'Nirukta' means 'etymology' and it explains the reason why a particular word has been used i.e., the meaning of usage. The only work which has survived as a specimen of this Vedanga 'etymology' is the Nirukta of Yaska. It is a commentary on *Nighantu* which is 'list of words' found in the Vedas. Tradition ascribes the Nighantu also to Yaska. The Nighantus are five lists of words, which are again divided into three sections. The first section consists of three lists, in which Vedic words are collected under certain main ideas. The second section contains a list of ambiguous and particularly difficult words of the Veda, while the third section gives a classification of the deities according to the three regions, earth, sky and heaven. Yaska explained these lists in the twelve books followed. The most interesting portion of the Nirukta is the discussion which covers the whole of the first book and a part of the second, as well as the seventh book, which was as an admirable introduction to the study of the Veda.

Yaska has mentioned a considerable number of important grammarians as his predecessors in the Nirukta such as Galava, Shakapuni, Katthakya.

Niruka is very important for several reasons. Firstly, it represents the type of the earliest classical style and in this respect stands by itself. Secondly, it is the oldest known attempt in the field of Vedic etymology. As regards the importance of the etymology Yaska himself says that without it the precise meanings of the Vedic stanzas cannot be understood

5. Chandah (Prosody or meter): Chandah studies the metrics and poetic structures of Vedic hymns. It focuses on the rhythm and meter of verses, which are essential for proper recitation and the aesthetic appreciation of Vedic literature.

Chandas Vedanga is regarded as the feet of the Veda Purusha. The body of the Vedas rests on the Chandas which are in the nature of feet. Each Mantra of the Veda has a special Chandas, just as it has a presiding Devata.

According to Nirukta the term Chandas is derived from the root Chad (to cover). Meter is called Chandas because it covers the sense of the Mantra. The Chandas is designed for the purpose of securing the proper reading and reciting of Vedic texts. The literature comprising this Vedanga on metrics is equally small.

The texts, dealing with Vedic meters, are as follows:

1. Rikpratishakhya
2. Shankhayana Shrauta-Sutra
3. Nidana-Sutra of Samaveda
4. Chandas-Sutras of Pingala

Each of them contains a section varying slightly from each other on Vedic meters.

6. Jyotisha (Astronomy and Astrology): Jyotisha encompasses the study of celestial bodies and their influence on human affairs. It includes calendar calculations, astrological predictions, and the timing of rituals based on astronomical observations.

The last Vedanga Jyotisha is called eye - the organ of sight, of the Veda Purusha. The object of Jyotisha Vedanga is not to teach astronomy, but to convey such knowledge of the heavenly bodies as is necessary for fixing the days and hours of the Vedic sacrifices. It gives some rules for calculating and fixing time for sacrifices. In the Brahmanas and Aranyakas, we find frequent allusions to astronomical subjects, and even in the hymns we find traces which indicate a certain advance in the observation of the moon.

It is unfortunate that there is no work available at present dealing with ancient Vedic astronomy (Jyotisha) in the Sutra style. Only we have a small text-book called Jyotisha of Vedic astronomy in verses in two recessions. Generally, Maharshi Lagadha is regarded author of this Vedanga Jyotisha. This is a very difficult text and, therefore, is not clear on several points to scholars even today. Later, we find many Sanskrit treatises on astronomy and mathematical calculations. Bhaskaracharya, Varahamihira and Aryabhatta are known ancient scholars conversant with these scientific subjects. The principles established by them are in use in the modern world.

Guru-Shishya Tradition:

The guru-shishya tradition, also known as guru-parampara, has been an integral part of Indian knowledge system for centuries. The origins of the guru-shishya tradition can be traced back to ancient India, where it was an essential aspect of the Vedic system of education. It was believed that the knowledge and wisdom of the guru could only be acquired through a personal and dedicated relationship between the guru (teacher) and shishya (disciple).

The guru would pass down his knowledge to the shishya through conversations, discussions, and demonstrations, and the shishya was expected to observe, absorb and implement the teachings. This method of learning was highly effective as it allowed for a deeper understanding of the subject, personalization of knowledge, and clarification of doubts. Furthermore, this tradition also emphasized the importance of the guru's guidance in not just acquiring knowledge but also in personal growth and character development.

In today's fast-paced world, where knowledge is easily accessible through various mediums, the guru-shishya tradition continues to hold immense relevance. It serves as a reminder of the

importance of a teacher in one's life and the value of personal guidance in the pursuit of knowledge.

Ayurveda:

Ayurveda is the ancient Indian system of medicine that emphasizes balance between the body, mind, and spirit. It is based on the concept of the doshas (Vata, Pitta, and Kapha) and focuses on diet, herbs, yoga, meditation, and lifestyle practices to maintain health and prevent disease.

Key texts: Charaka Samhita, Sushruta Samhita, and Ashtanga Hridayam.

Mathematics and Astronomy:

- Indian Mathematics made significant early contributions to concepts like the decimal system, the concept of zero, algebra, and trigonometry.
- Notable mathematicians such as Aryabhata, Brahmagupta, and Bhaskara II made substantial advances in calculating the value of pi and solving quadratic equations.
- Astronomy in India, exemplified by texts like the Surya Siddhanta, was advanced and included methods for calculating time, the movement of planets, and the structure of the cosmos.

Jainism and Buddhist Knowledge Systems:

- Both Jainism and Buddhism, though originating in India, provide distinct philosophical traditions.
- Jainism emphasizes non-violence (Ahimsa), truth, and asceticism, with a highly systematic approach to logic, metaphysics, and ethics.
- Buddhism, founded by Siddhartha Gautama, addresses the nature of suffering, the path to enlightenment, and the doctrines of impermanence and non-self. Its philosophical system includes teachings like the Four Noble Truths (dukkha, samudaya, nirodha, magga) and Eightfold Path (sammā diṭṭhi, sammā saṅkappa, sammā vācā, sammā kammanta, sammā ājīva, sammā vāyāma, sammā sati, sammā samādhi).

#####

Unit 3

Indian Mathematics

Numbers, fractions and geometry in the Vedas. Decimal nomenclature of numbers in the Vedas. Zero and Infinity. The development of the decimal place value system which resulted in a simplification of all arithmetical operations. Linguistic representation of numbers. Important texts of Indian mathematics. Brief introduction to the development of algebra, trigonometry and calculus.

1. Introduction:

Indian Mathematics has a rich and deep history, deeply embedded in the Indian Knowledge System (IKS). The Indian subcontinent has contributed significantly to the development of mathematics, and many concepts from ancient Indian mathematics have influenced global mathematical thought.

Here are some key areas where Indian mathematics has made important contributions:

1. Numeral System and the Concept of Zero:

- a) Decimal System (Place Value System): The Indian mathematician Aryabhata (5th century CE) and later Brahmagupta (7th century CE) were instrumental in formalizing the decimal system, which uses the place value of digits to represent numbers. This system is now universally used.
- b) Zero: The concept of zero as a number and its symbolic representation (a small dot or a blank space) was first explicitly used in India. Brahmagupta's work in 628 CE described the rules for arithmetic operations involving zero.

2. Algebra:

Brahmagupta's work in algebra was foundational. In his book "Brahmasphutasiddhanta," he discussed solutions to quadratic equations, the concept of negative numbers, and the rules for addition and subtraction of zero. The Bakhshali Manuscript (circa 300 CE) provides an early example of algebraic methods and uses the concept of indeterminate equations.

3. Trigonometry:

Indian mathematicians were among the first to develop trigonometric concepts. Aryabhata introduced sine (known as "jya") and cosine (known as "kojya") in the context of astronomy. Bhaskara I and Bhaskara II expanded these ideas, and Bhaskara II's "Lilavati" and "Bijaganita" provided deep insights into both algebra and trigonometry.

4. Pi (π) and its Approximation:

Indian scholars like Aryabhata and Madhava of Sangamagrama (14th century) were among the first to approximate the value of π . Aryabhata calculated the value of π as 3.1416, which is remarkably close to the actual value. Madhava's series for π , the Madhava-Leibniz series, is a precursor to modern infinite series expansions used in calculus.

5. Geometry:

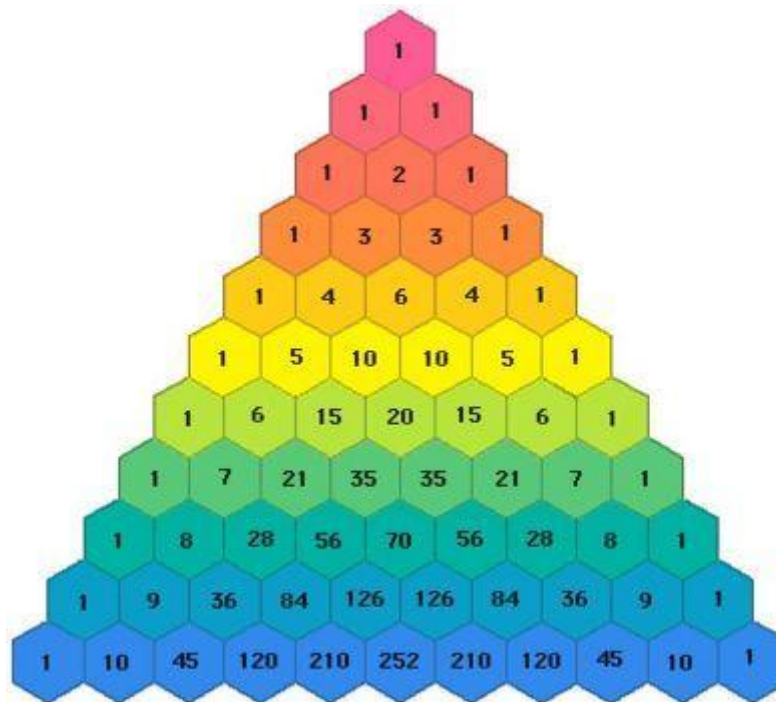
Sulba Sutras (c. 800 BCE) provide the earliest Indian texts on geometry. They primarily focus on the construction of altars but contain geometric principles and methods, including the Pythagorean theorem, well before it was known in the West. Brahmagupta also worked on geometric properties, describing various theorems related to triangles, circles, and the area of a cyclic quadrilateral.

6. Calculus:

The concept of infinite series and calculus was developed in India much before the European Renaissance. Madhava and his followers (the Kerala School of Mathematics) developed infinite series expansions for trigonometric functions and discovered methods that resemble modern calculus. The work of Nilakantha, Jyesthadeva, and others in the 14th-15th centuries anticipated key aspects of calculus, such as differentiation and integration.

7. Combinatorics and Permutation Theory:

The ancient Jaina mathematicians of India contributed to combinatorics, studying combinations and permutations. The concept of binomial coefficients was already in use in ancient India, particularly in the form of “Pascal’s Triangle”, which was known to Indian mathematicians long before Blaise Pascal.



Pascal's Triangle

8. Mathematics in Astronomy:

Indian mathematics was closely tied to astronomy, as the need to calculate planetary positions and predict eclipses required advanced mathematical techniques. Aryabhata, Varahamihira, and Bhaskara II made major contributions to both fields, utilizing mathematics for celestial mechanics and timekeeping.

9. The Indian Mathematical Tradition:

India had a long-standing tradition of mathematical exploration. From Vedic texts to the works of medieval scholars, mathematics was integrated with other domains like philosophy, religion, and astronomy. The Vedas themselves contain numbers, sequences, and basic arithmetic, and the later Shulba Sutras contain geometrical principles.

10. Indian Mathematical Texts:

Some significant mathematical texts from ancient India include:

- ✓ Aryabhatiya by Aryabhata
- ✓ Brahmasphutasiddhanta by Brahmagupta
- ✓ Lilavati and Bijaganita by Bhaskara II
- ✓ Yuktibhāṣā by Jyesthadeva (14th century)

The contributions of Indian mathematicians to global mathematical thought were vast and influential. Many of the concepts we use today—such as the decimal system, zero, algebra, trigonometry, and calculus—have roots in the Indian mathematical tradition.

Indian mathematics was both practical and deeply theoretical, with applications in astronomy, engineering, and architecture. The work of ancient and medieval Indian scholars continues to be a source of inspiration for contemporary mathematics.

The Vedas, the oldest sacred texts of Hinduism, contain many references to numbers, fractions, and geometrical concepts, although these are not always presented in the way that modern mathematics would define or categorise them. The Vedic texts, primarily composed of hymns, prayers, rituals, and philosophical discourses, are deeply intertwined with cosmological ideas, and often numbers and geometry serve symbolic, metaphysical, and ritualistic purposes rather than being part of a systematic mathematical framework. Still, there are a number of significant

mathematical concepts found in the Vedas and related texts, especially in relation to cosmology, time, ritual, and architecture.

1.1 Numbers and Counting in the Vedas

The Vedic texts use numbers in various ways, often in the context of ritual practices, cosmology, and the measurement of time:

- ✓ **Cosmological Numbers:** The Vedas describe cycles of time, such as Yugas (ages), which are measured in thousands, hundreds of thousands, and millions of years. The concept of a Mahakalpa (a "great age" or "cosmic cycle") is especially prominent.
- ✓ **Kalpa:** A Kalpa (one day of Brahma, the creator) consists of 4.32 billion years. This astronomical timekeeping system is an example of how large numbers were applied to the understanding of cosmic cycles.
- ✓ **Ritual Numbers:** Rituals in the Vedas often use specific numbers, especially in connection with offerings, repetitions of mantras, or the performance of sacrifices. For instance, the Soma sacrifice in the Rigveda involves certain sets of repetitions and groupings, such as 3, 5, 7, or 12, which have symbolic significance in Vedic cosmology.
- ✓ **The Number 1,000:** The number 1,000 is particularly important in the Vedic texts, often signifying completeness or the totality of a cycle. For example, the term "Sahasra" (meaning thousand) appears frequently in the Rigveda, such as in the description of the Sahasranama, or the "Thousand Names" of deities like Vishnu, which suggests an idea of completeness or infinity.

A. Concept of Numbers

- ✓ The Vedic people used numbers for rituals, astronomy, trade, and measurements.
- ✓ Numbers were not just mathematical but had philosophical and symbolic meanings.
- ✓ Example:
 - 1 represented unity (Atman or Brahman)
 - 2 represented duality (day-night, male-female)
 - 3 represented the three worlds (Earth, Heaven, Atmosphere)
 - 7 represented spiritual perfection (7 days, 7 sages, etc.)

B. Large Numbers

- ✓ Vedic texts show familiarity with very large numbers, showing the advanced nature of Indian mathematics.
- ✓ Example from *Yajurveda* and *Taittiriya Samhita*:
Numbers up to 10^{12} (one trillion) are mentioned!
 - 1 → Eka
 - 10 → Dasa
 - 100 → Shata
 - 1,000 → Sahasra
 - 10,000 → Ayuta
 - 100,000 → Laksha
 - 10 million → Koti (1 crores)
 - And higher powers continue: arbuda, abja, kharva, nikharva, mahapadma, shankha, jaladhi, antya, madhya, parardha (10^8 to 10^{17})

🌈 **Fun fact:** This early use of powers of ten shows that India had developed the decimal place-value system much before others.

1.2 Fractions in the Vedas

A. Early Understanding

Fractions were known and used, especially in Sulba Sutras (texts dealing with geometry for fire altars). They used fractions for measurements and constructions.

Example:

- $\frac{1}{2} \rightarrow \text{ardha}$

- $\frac{1}{4} \rightarrow pāda$
- $\frac{3}{4} \rightarrow tripāda$
- $\frac{1}{8} \rightarrow aṣṭamāṃśa$

B. Practical Usage:

Fractions were used in:

- Dividing sacrificial grounds
- Sharing goods
- Astronomical calculations.

Fractions appear in the Vedic texts, especially in the context of measuring time, proportions in rituals, or the division of offerings:

- ✓ **Time Fractions:** The Vedas mention various fractional parts of time, such as the division of the day (for example, parts of the day are sometimes referred to in fractions like $\frac{1}{4}$ or $\frac{1}{3}$). The Puranas, which are later texts related to the Vedic tradition, elaborate on these fractional time periods, such as the division of the day into Yamas (periods), and the measurement of cosmic time in smaller fractions like Truti ($\frac{1}{75}$ of a second), Nimesha (the twinkling of an eye), and so on.
- ✓ **Ritual Fractions:** Ritual sacrifices might also involve fractional divisions of offerings, such as the division of the Soma drink or the distribution of sacred fire. These fractions were likely symbolic, representing cosmic or divine harmony.

1.3 Geometry in the Vedas

A. Geometry in the Śulba Sūtras

- ✓ *Śulba* means “cord” — geometry was done using **measuring ropes**.
- ✓ These texts (especially by Baudhāyana, Āpastamba, and Kātyāyana) deal with **construction of altars (yajnas)** of specific shapes and sizes.

B. Key Geometrical Concepts

i. Right-angled Triangle:

- The *Baudhāyana Śulba Sūtra* mentions:
“The diagonal of a rectangle produces both areas which the two sides produce separately.”
→ This is essentially the **Pythagoras Theorem!**

$$a^2 + b^2 = c^2$$

ii. Geometric Constructions:

- Methods to construct:
 - Squares with equal area as a rectangle
 - Circles with equal area as a square (approximation of π)
 - Example: *To make a square equal in area to a circle, increase the diameter by $\frac{1}{8}$ and draw the square on that.*

C. Units of Measurement:

- ✓ *Angula* (finger width), *hasta* (hand span), *purusha* (height of a man), etc.

Geometry in the Vedas is less formalized than in later Indian mathematics (e.g., in the works of scholars like Aryabhata and Brahmagupta), but some geometrical concepts can be inferred from the texts, especially in the context of architecture, ritual, and cosmology.

- ✓ **Vedic Fire Altars (Yajnas):** One of the most direct references to geometry in the Vedic period is the design of the yajna (sacrificial) altar. The Yajurveda describes the construction of fire altars, which had specific geometrical shapes and dimensions. For instance, the altar for the Agnihotra (a fire ritual) is described as being square or rectangular, with specific proportions of length, breadth, and height. These designs were often seen as corresponding to cosmic principles, with the altar being a microcosm of the universe.
- ✓ **Vastu Shastra:** The art of architectural design, Vastu Shastra, which emerges in later texts, draws heavily from the Vedic tradition and includes geometric principles for the construction

of temples, homes, and altars. While the formalization of Vastu Shastra came after the Vedic period, it is rooted in Vedic concepts of sacred geometry. The design of sacred spaces was often thought to reflect the structure of the universe, and geometric shapes like the “mandala” (a circular design representing the cosmos) played a significant role in rituals.

- ✓ **Sacred Geometry:** The mandala, a geometric design representing the universe, is another example of geometry in the Vedic tradition. Though mandalas are more commonly associated with later Hinduism and Buddhism, they have their origins in Vedic symbolism. The mandala represents the cosmic order and is used as a tool in rituals to invoke cosmic harmony.

1.4 Symbolic Use of Numbers and Geometry in Vedic Cosmology

The Vedic texts often employ numbers and geometric shapes symbolically:

- ✓ **The Cosmic Order:** The number 3 is frequently associated with fundamental concepts in Vedic cosmology: the three main deities (Brahma, Vishnu, and Shiva), the three worlds (Earth, Heaven, and the Netherworld), and the three gunas (qualities of nature: sattva, rajas, and tamas). The number 7 is also important, representing completeness or perfection, as seen in the ‘Seven Sages’ (the Saptarishi), the seven circles of the universe, and the seven chakras in later traditions.
- ✓ **The "Primal Geometry":** Some scholars believe that the geometric symbols used in Vedic rituals, such as the Shri Yantra (a geometric representation of the divine feminine energy), point to an early form of sacred geometry that underlies the structure of the cosmos and the human body. These symbols suggest an intimate relationship between the microcosm (human) and the macrocosm (universe).

1.5 Mathematical Knowledge beyond the Vedas

Though the Vedas themselves do not focus on formal mathematics as we understand it today, they laid the groundwork for later developments. Texts such as the Sulba Sutras, which are part of the Kalpa (ritual) literature, contain specific geometrical and numerical instructions for constructing altars used in Vedic rituals. The Sulba Sutras describe principles of geometry, including the Pythagorean Theorem, and are some of the earliest recorded examples of geometric and arithmetic knowledge in Indian tradition.

Decimal Nomenclature of Numbers in the Vedas:

a. Decimal System

- The **base-10 system** was known and used in Vedic times.
- The names of numbers were arranged in **powers of 10**.
- Example:

10^1 – *Dasa (Ten)*

10^2 – *Shata (Hundred)*

10^3 – *Sahasra (Thousand)*

10^4 – *Ayuta (Ten Thousand)*

10^5 – *Laksha (Lakh)*

10^6 – *Prayuta (Ten Lakh)*

10^7 – *Koti (1cr)*

10^8 – *Arbuda (10cr)*

10^9 – *Abja (100cr)*

10^{10} – *Kharva (1000cr)*

10^{11} – *Nikharva* (10,000cr)

10^{12} – *Mahāpadma*, One lakh crore: literally great lotus.

Śaṅkha (10^{13})-- Ten lakh crore-- Literally “conch”

Jaladhi (10^{14})-- One crore crore-- Literally “ocean”

Antya (10^{15})-- Ten crore crore-- “Endless” or “final”

Madhya (10^{16})-- Hundred crore crore-- “Middle”

Parārdha (10^{17})-- Thousand crore crore-- Beyond measure” — the highest traditional term

→ This shows the **hierarchical and positional understanding** of numbers — the foundation of the **decimal place-value system** that we use today.

Conclusion:

- Vedic mathematics is **not just about arithmetic** — it shows how deeply mathematics was integrated into **religion, architecture, and philosophy**.
- The **decimal system and geometric principles** developed in the Vedic period became the **foundation of modern mathematics**.

Summary of Contributions:

- | | |
|--------------------|---|
| • Area | • Vedic Contribution |
| • Numbers | • Concept of large numbers, decimal place-value system |
| • Fractions | • Used for measurements and altar constructions |
| • Geometry | • Knowledge of shapes, right triangles, area equivalence |
| • Notation | • Systematic naming up to very large numbers |
| • Legacy | • Foundation for later Indian mathematics (Aryabhata, Brahmagupta.) |

Linguistic Representation of Numbers:

Number	Sanskrit Name	Transliteration
1	एक	Eka
2	द्वि / द्वौ	Dvi
3	त्रि	Tri
4	चतुर्	Chatur
5	पञ्च	Pancha
6	षट्	Shat
7	सप्त	Sapta
8	अष्ट	Ashta
9	नव	Nava
10	दश	Dasha
11	एकादश	Ekādasha

12	द्वादश	Dvādasha
13	त्रयोदश	Trayodasha
14	चतुर्दश	Chaturdasha
15	पञ्चदश	Panchadasha
16	षोडश	Shodasha
17	सप्तदश	Saptadasha
18	अष्टादश	Ashtādasha
19	नवदश	Navadasha
20	विंशति	Vimshati

Quick Pattern to Remember:

- 11–19 are formed as:
(**Number name**) + **dasha** → e.g., *Eka + dasha = Ekādasha* (11).
- 20 is **Vimshati**.
- This system continues ($30 = \text{Trimshat}$, $40 = \text{Chatvārimshat}$, etc.) in higher numbers.

Introduction:

- India has one of the **oldest and most continuous mathematical traditions** in the world.
- Early mathematics is found in the **Vedas**, **Śulba Sūtras**, and later in classical texts by **Aryabhata**, **Brahmagupta**, **Bhaskara**, **Madhava**, and others.
- These contributions laid the foundation of **modern arithmetic, algebra, trigonometry, and calculus**.

Numbers in the Vedas:

- Numbers were used in **rituals, astronomy, trade, and geometry**.
- Vedic people had knowledge of **small and extremely large numbers**.
- Example:
 - 1 (Eka), 10 (Dasa), 100 (Shata), 1000 (Sahasra), 10^6 (Prayuta), 10^7 (Koti), up to 10^{12} (Mahapadma).
- This system shows the **use of powers of 10**, the **basis of the decimal system**.

Zero (Śūnya) and Infinity (Ananta):

a. The Concept of Zero:

- The idea of “**nothingness**” or “**void**” (**Śūnya**) comes from Indian philosophy and mathematics.
- Zero was first used as a **placeholder** (to show empty positions in numbers).
- Later, it was treated as a **number itself**, allowing new kinds of arithmetic operations.

🔑 Key contributors:

- Brahmagupta (7th century CE)** was the first to define operations with zero:
 - $a + 0 = a$
 - $a - 0 = a$
 - $a \times 0 = 0$
 - $\frac{a}{0} \rightarrow$ undefined or infinity concept

b. The Concept of Infinity:

- Indian mathematicians recognized **infinity (Ananta)** as both a **mathematical** and **philosophical** idea.
- Example:
 - “If you remove a part from infinity, what remains is still infinity.”
- Used in discussions on **geometry, astronomy, and philosophy (Upanishads)**.

The Decimal Place-Value System:

a. Origin

- The **place-value system** originated in India around the **1st–4th century CE**.
- It means the **value of a digit depends on its position** (ones, tens, hundreds, etc.).
 - Example: $345 = (3 \times 100) + (4 \times 10) + (5 \times 1)$
- Combined with **zero**, this made arithmetic operations simple and efficient.

b. Importance

- Simplified **addition, subtraction, multiplication, and division**.
 - Easier to **record and compute large numbers**.
 - Spread to the world through **Arabic scholars**, hence the term “**Hindu–Arabic numeral system**.”
-

Linguistic Representation of Numbers:

- Numbers were often written in **words or poetic form** to preserve knowledge orally.
 - Each number had a **word symbol** in Sanskrit.
 - Example:
 - $1 \rightarrow eka$, $2 \rightarrow dvi$, $3 \rightarrow tri$, $4 \rightarrow chatur$, $5 \rightarrow panch$, $10 \rightarrow dasha$
 - Later, the **Katapayadi system** (used in Kerala School) used **letters to represent numbers**.
 - Example: certain consonants represented digits 1–9, vowels were ignored.
 - Used in **astronomical texts** to encode numerical data in verses.
-

Important Texts of Indian Mathematics:

Period	Text	Author	Key Contributions
Vedic (1500–500 BCE)	<i>Sulba Sutras</i>	Baudhāyana, Āpastamba, Kātyāyana	Geometry, altar constructions, Pythagoras theorem
Classical (400–1200 CE)	<i>Āryabhaṭīya</i>	Aryabhata (476 CE)	Place-value system, zero, trigonometric functions (sine, cosine)
	<i>Brāhmasphuṭasiddhānta</i>	Brahmagupta (7th century)	Rules for zero, negative numbers, quadratic equations
	<i>Līlāvati</i>	Bhaskara II (12th century)	Arithmetic, algebra, geometry in verse form
	<i>Siddhānta Shiromani</i>	Bhaskara II	Trigonometry, planetary motion
Medieval (1300–1600 CE)	<i>Yuktibhāṣā</i>	Jyeshthadeva (Kerala School)	Early form of calculus, infinite series

Development of Algebra, Trigonometry, and Calculus

a. Algebra (Bijaganita)

- Originated from the need to solve **unknown quantities** in equations.
- **Brahmagupta** introduced:

- Rules for **positive and negative numbers**
- **Quadratic equations**
- Symbolic representation of unknowns
- Later, **Bhaskara II** refined these methods and gave solutions for **indeterminate equations**.

b. Trigonometry (Trikonamiti)

- Developed first for **astronomical calculations**.
- **Aryabhata** introduced the concept of “**sine**” (**jya**) and “**cosine**” (**kojya**).
- He calculated:
 - Values of sine for different angles.
 - Formula for approximate value of π ($\pi \approx 3.1416$).
- **Bhaskara II** later gave more accurate trigonometric identities.

c. Calculus (Integration and Infinite Series)

- Developed by the **Kerala School of Mathematics (14th–16th century)**.
- **Madhava of Sangamagrama** discovered:
 - **Infinite series** for π , sine, and cosine (long before Newton or Leibniz).
 - Concepts similar to **differentiation and integration**.
- These ideas mark the **beginning of modern calculus**.

Concept

Contribution

Zero (Śūnya) Treated as a number; rules by Brahmagupta

Infinity (Ananta) Recognized in math and philosophy

Decimal System Place-value with base 10 and zero

Fractions Used in geometry and altar designs

Algebra Rules for equations and unknowns

Trigonometry Sine, cosine, π from Aryabhata

Calculus Infinite series by Madhava

Texts Sulba Sutras, Aryabhata, Lilavati, Yuktibhāṣā

Conclusion

- Indian mathematicians gave the world **the concept of zero, the decimal system, and the foundations of algebra, trigonometry, and calculus**.

- Their work shows how **philosophy, science, and mathematics** were deeply connected in ancient India.
- These ideas travelled across civilizations and became the **core of modern mathematics**.

Śulba Sūtras — Meaning and Importance

♦ Meaning:

- The word **Śulba** means “*measuring cord*” or “*rope*”.
- **Sūtra** means “*rule*” or “*aphorism*”.
- So, **Śulba Sūtras** literally means “**rules of the cord**”, referring to using a rope to measure and construct altars and ritual spaces with accurate geometry.

What They Are:

- The **Śulba Sūtras** are **ancient geometry manuals** that explain how to:
 - Construct **fire altars (vedi)** of precise shapes and sizes.
 - Maintain **equal area** when changing altar shapes (e.g., square to circle).
 - Use **geometric principles** for accurate measurements using a rope.

Time Period:

- Believed to have been composed between **800 BCE and 300 BCE** (some parts may be even older).
- They are among the **oldest known mathematical texts in the world**.

Famous Śulba Sūtra Authors:

Author	Approx. Period	Notable Features
Baudhāyana	~800 BCE	Contains early statement of the Pythagorean theorem long before Pythagoras.
Āpastamba	~600 BCE	Gives accurate geometric constructions and value of $\sqrt{2}$ (≈ 1.4142).
Kātyāyana	~300 BCE	Refines earlier methods.
Mānava	~500 BCE	Contains additional altar designs.

Mathematical Contributions:

The **Śulba Sūtras** show that ancient Indians had deep knowledge of:

- **Geometry** – constructing squares, rectangles, circles, triangles.
- **The Pythagorean theorem** — stated as:

“The diagonal of a rectangle produces both areas which the two sides produce separately.”

- **Approximation of $\sqrt{2}$** — very close to the modern value.
- **Concept of irrational numbers and area equivalence.**

#####

Syllabus

Unit 4:

- **Indian Astronomy: Ancient record of the observation of the motion of celestial bodies in the Vedic corpus.**
- **Sun, Moon, Nakshatra and Graha.**
- **Astronomy as the science of determination of time, place and direction by observing the motion of the celestial bodies**
- **The motion of the Sun and Moon.**
- **Motion of equinox and solstices**
- **Elements of Indian calendar systems.**
- **Basic ideas of the planetary model of Aryabhatta and its revision by Nilkantha**

Indian Astronomy: Ancient record of the observation of the motion of celestial bodies in the Vedic corpus.

- Indian astronomy—Jyotiṣa—has its earliest foundations in the Vedic literature, which contains some of the world's oldest records of systematic observation of celestial bodies. These texts, dating from roughly 1500–500 BCE, reflect an advanced understanding of the sky, timekeeping, seasons, and cosmic order.
- Indian astronomy has its roots in the Vedas (1500 BCE), the oldest sacred texts of Hinduism.
- Vedas as the Earliest Astronomical Records

R̥gveda (1500–1200 BCE)

- The oldest Vedic text contains numerous hymns referencing celestial phenomena.
- Mentions of Sun (Sūrya), Moon (Candra), Dawn (Uṣas), and Nakṣatras (stars) show a deep familiarity with the sky.
- Observations include:
 - The daily motion of the Sun across the sky.
 - The phases of the Moon.
 - The understanding that the year is determined by solar movement.
 - Stars rising and setting at specific seasons.

Yajurveda (1200–1000 BCE)

- Contains more developed astronomical references.
- Lists 27 (sometimes 28) Nakṣatras, the earliest known systematic star calendar.
- Describes:
 - The solar year of 365 days.
 - The concept of intercalary months (Adhikamāsa) to harmonize lunar and solar calendars.
 - The movement of the Sun through different directions.

Atharvaveda

- Provides references to eclipses (Grahana), indicating observational awareness.
- Mentions of seven planets visible to the naked eye.

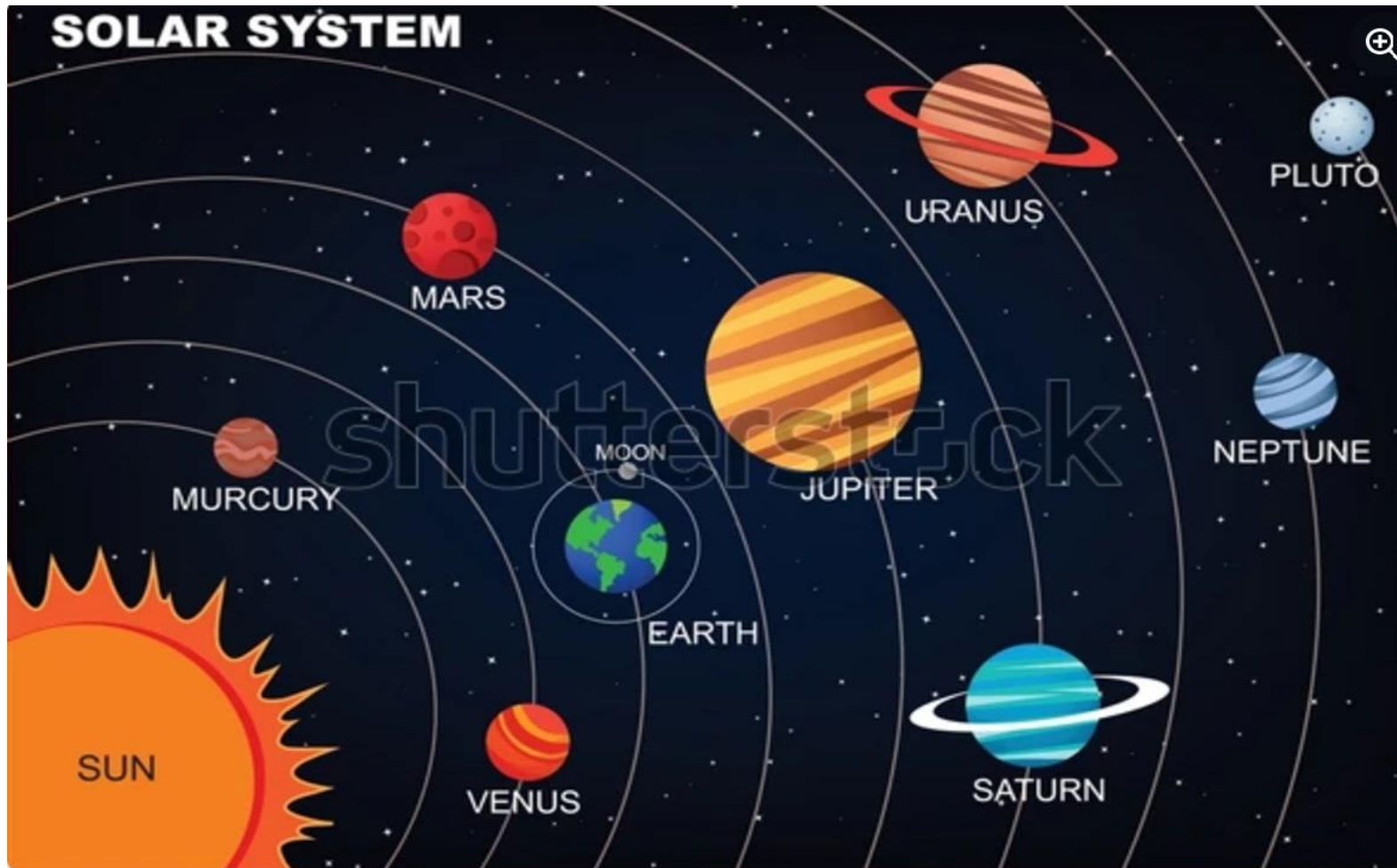
Sāmaveda

-
- Mostly musical, but ritual timings depend on stellar positions, preserving astronomical knowledge.

Sun, Moon, Nakshatra and Graha.

- Ancient Indian astronomers made significant progress in understanding planetary motion.
- * They developed mathematical models to describe elliptical planetary paths and predict positions.
- * The Siddhantas, texts on astronomy, provided detailed information on planetary motion and positions.
- * Nakshatras, or 27 lunar mansions, were important in astronomy and astrology, guiding cultural and religious practices.

Sun and Moon



Sun (Sūrya)

- Daily east-to-west motion.
- Northward and southward movement (precursor to Uttarāyaṇa and Dakṣiṇāyana).
- Seasonal change as the basis of the year.

Moon (Candra)

- observed phases meticulously .
- Lunar mansions indicate continuous tracking of its 27 .3-day sidereal cycle.



What is a Nakshatra:

- * The word "Nakshatra" comes from "Naksha" (map) and "Tara" (star).
- * Mentioned in the Rig Veda, Yajurveda, and Atharva Veda, it refers to the mapping of stars.
- * In Indian astrology, Nakshatras are constellations or star groups affecting a person's life from birth.
- * Ancient Indian astronomy divides the ecliptic into 27 nakshatras.
- * The Vedanga Jyotisha (final centuries BCE) contains the earliest list of these nakshatras.

Nakshatra

- Ashvini
- Bharani
- Krittika
- Rohini
- Mrigashirsha
- Ardra
- Punarvasu
- Pushya
- Ashlesha
- Magha
- Purva Phalguni
- Uttara Phalguni
- Hasta
- Chitra
- Swati
- Vishakha
- Anuradha
- Jyeshtha
- Mula
- Purva Ashadha
- Uttara Ashadha
- Shravana
- Dhanishtha
- Shatabhisha
- Purva Bhadrapada
- Uttara Bhadrapada
- Revati

Nakṣatra System:

Observational Significance

Each Nakṣatra corresponds to a recognizable star or constellation.

Used for:

Ritual timings

Agriculture

Navigation

Seasonal prediction

are mentioned in Vedic Texts

Planets (Grahas)

Venus (Śukra) mentioned as morning and evening star.

- Jupiter (Bṛhaspati) connected to 12-year cycles.
- Saturn (Śani), Mars (Maṅgala), and Mercury (Budha) appear as sky markers.

Eclipses

- Words such as Svarbhānu and Rāhu in Vedic mythology metaphorically describe solar and lunar eclipse.
- Indicate an understanding that eclipses are periodic, observable celestial events.

Graha



Astronomy as the science of determination of time, place and direction by observing the motion of the celestial bodies

Astronomy is one of the earliest sciences developed by humankind. By carefully observing the motion of celestial bodies such as the Sun, Moon, planets, and stars, ancient scholars were able to determine the fundamental coordinates of life:

Time:

Celestial cycles helped measure days, months, seasons, and years. Sunrise, sunset, lunar phases, and solstices became markers of timekeeping.

Time Measurement

- **Precise terms:**
- Ahōrātra (day-night cycle)
- Tithi (lunar day)
- Māsa (month)
- Rtu (season)
- Samvatsara (year)

Place

The positions of stars and the Sun enabled the determination of latitude and longitude, helping identify one's location on Earth.

Direction:

Fixed celestial markers like the Pole Star provided reliable orientation, guiding travelers, sailors, and explorers.

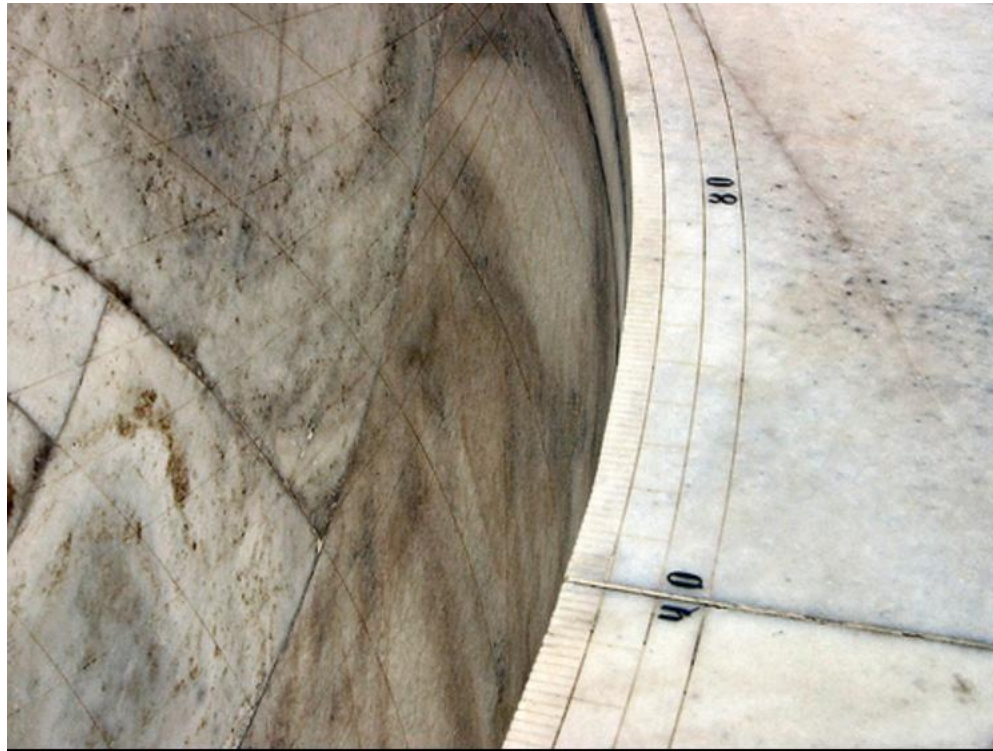
- * Ancient Indian astronomers developed instruments to aid celestial observations and calculations.
- * Yantra: geometrical devices used to measure angles and determine celestial positions.
- * Ram Yantra and Digamsha Yantra: examples of precise observational instruments.
- * Gnomon: a vertical rod used to measure the Sun's height and calculate time and latitude.
- * Astrolabe: a complex instrument for precise measurements of stars and planets, aiding in predicting celestial events like eclipses.

The Jantar Mantar, Jaipur



- * Jantar Mantar, Jaipur – an astronomical observatory built in the early 18th century by Maharaja Sawai Jai Singh II.
- * Comprises about 20 fixed masonry instruments used for naked-eye observation of celestial positions.
- * Instruments are monumental versions of traditional designs, each with unique architectural features.
- * Reflects the astronomical expertise and cosmological vision of the scholarly Mughal-era court.
- * It is the largest and best-preserved set of monumental astronomical instruments in India.
- * Embodies architectural and instrumental innovations for accurate celestial observation.
- * Based on the Ptolemaic system of positional astronomy, shared across ancient civilizations.
- * Contributed to refining astronomical tables (Zij) through precise observations.
- * Served as a center of cross-cultural scientific exchange under Jai Singh II's patronage.
- * Functioned as a symbol of royal authority, linking political power, science, and religion through control of time and astrological forecasting.

Samrat yantra with divisions of hours, minutes and seconds





Sun Dial



Chakra Yantra

- * Chakra Yantra consists of four hemispherical open disks used to determine the Sun's declination at four specific times of the day.
- * It functions more as a calculator than an observational instrument.
- * A central post casts a shadow, allowing readings to be taken.
- * The data corresponds to 12 noon at four observatories: Zurich (Switzerland), Greenwich (UK), Saitchen (Pacific Ocean), and Notke (Japan).
- * The instrument provides the local time in different parts of the world.
- * Nearby is the Kapali Yantra, seen as two semicircular structures on either side of the Chakra Yantra.



- * The southern dial of the smaller Samrat Yantra measures the Sun's height in the sky.
- * It bears Sanskrit inscriptions related to the conservation and renovation of the Narivalaya Yantra.
- * The inscription begins with a mention of Lord Ganesh.
- * It explains the purpose of the astrological instrument. ©
- * The script acknowledges the earlier builder's thirst for knowledge.
- * It provides details of the instrument's renovation.



Iron post

- * The iron post is at the center of a Rama Yantra, with triangular spokes fanning out to the walls.
- * The instrument is simple: readings are taken from numbers inscribed on the floor, ring on top, and walls.
- * Two Rama Yantras work together; when the shadow misses the spokes on one, it can be observed on the other.



- * The smaller Samrat Yantra, called Laghu Samrat Yantra, is near the entrance of Jantar Mantar.
- * The larger Samrat Yantra, located diagonally opposite, works on the same principle but on a bigger scale.
- * The triangular wall of each instrument aligns with Jaipur's north-south axis.
- * The shadow falling on the hemispherical curves indicates the time in Jaipur.
- * The larger Yantra has an accuracy of 2 seconds; the smaller one, 20 seconds.
- * The smaller instrument may have served as a test model for the larger one.
- * Despite its simplicity, it is fascinating to observe the Sun's shadow moving along the instrument on a clear day.

Sun Temple

- * Ancient Indian astronomy influenced temple architecture to align celestial events with structures.
- * The Sun Temple at Konark is oriented east to capture the first rays of the Sun.
- * At Konark, the Sun's rays fall on the image of Surya during the equinoxes.
- * While the Shikaras no longer survive, the Torana, stepwell, pillars, Garbhagriha, and tank in front still remain.



The motion of the Sun and Moon.

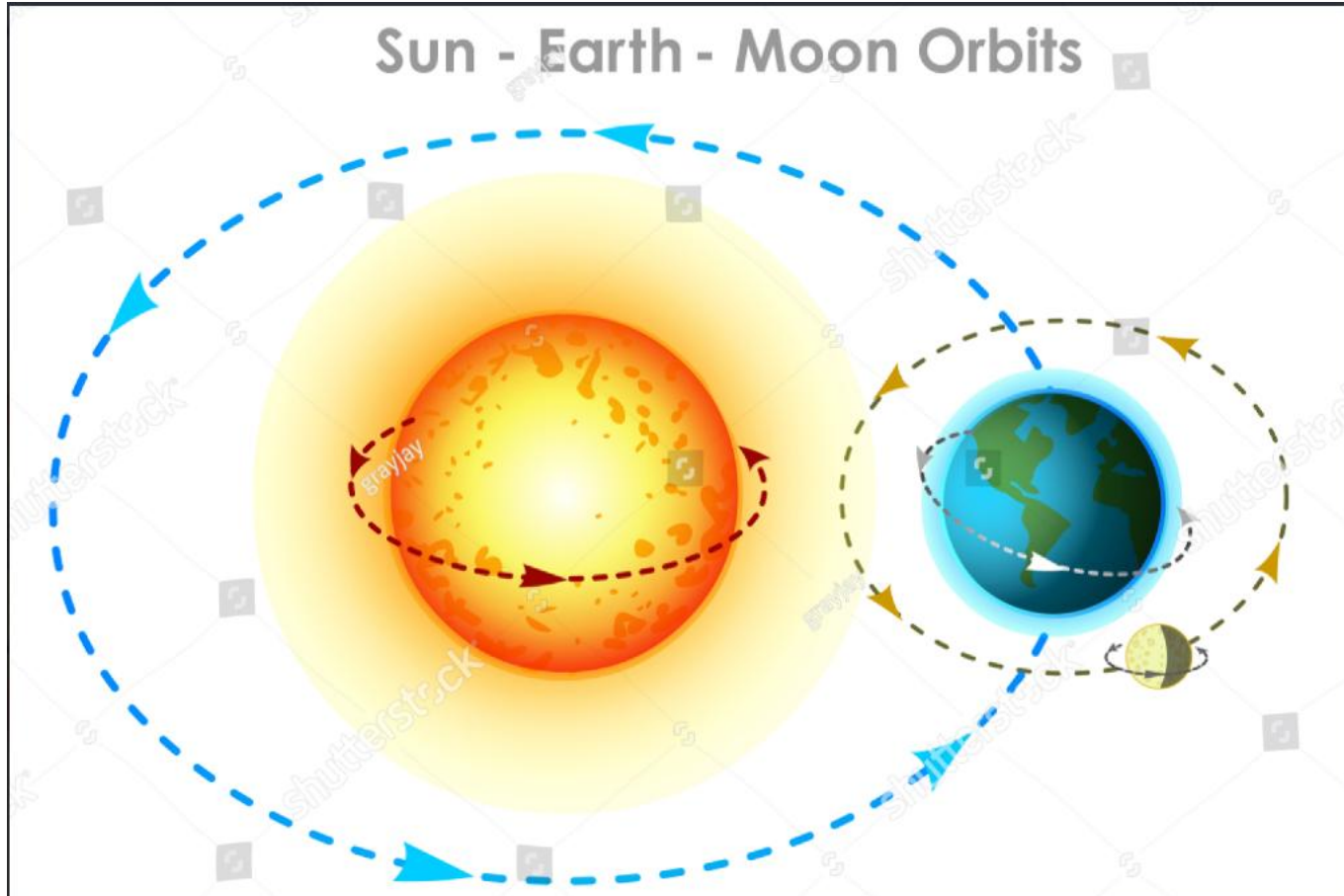
- Ancient Indian astronomers studied lunar and solar phenomena closely.
- The Tithi System, a lunar calendar based on the moon's phases, helped determine auspicious dates for rituals and agriculture.
- The Surya Siddhanta described solar eclipses, their causes, periodicity, and prediction methods.
- Ancient Indians understood the celestial mechanics behind eclipses.
- Brahmagupta (7th century CE) in the Brahmasphutasiddhanta detailed planetary motion, predicted eclipses, and provided formulas to calculate planetary positions.

- The Sun rises in the east, moves across the sky and sets in the west. Its path is called the ecliptic, and it rotates once every 25 days.
- The Earth rotates once every 24 hours, causing day and night, and orbits the Sun once every $365\frac{1}{4}$ days.
- The Moon orbits the Earth every $27\frac{1}{2}$ days, but takes about $29\frac{1}{2}$ days to return to the same position in the sky due to Earth's motion.
- The Moon's orbit is tilted 5° relative to Earth's orbit, and it is tidally locked, so only one side of the moon faces Earth.
- Moon phases result from the relative positions of the Earth, Moon, and Sun.
- Earth's axis is tilted 23.5° , causing seasons; during the summer solstice sunlight hits more directly, and during the winter solstice, at a shallower angle.

Our Earth

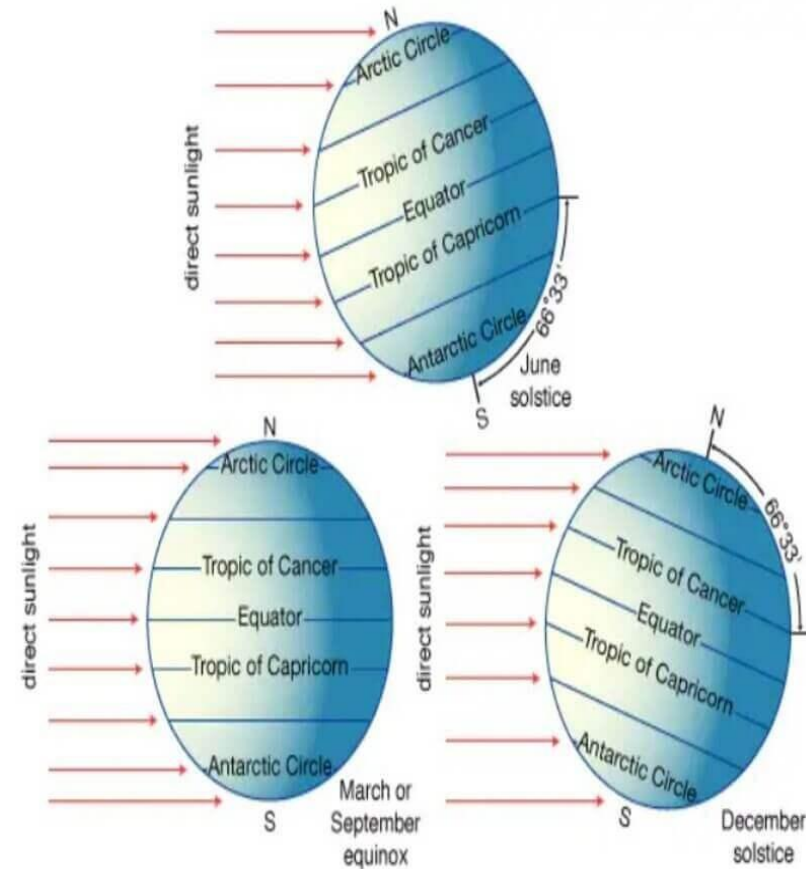
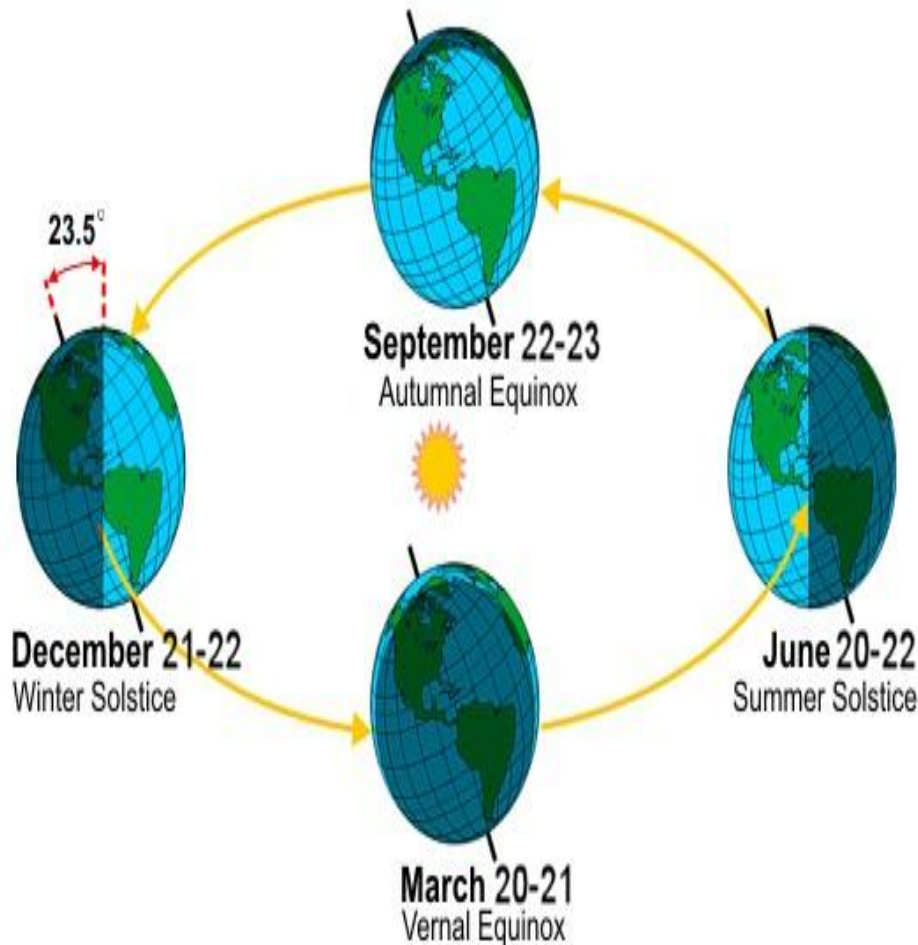


Sun - Earth - Moon Orbits



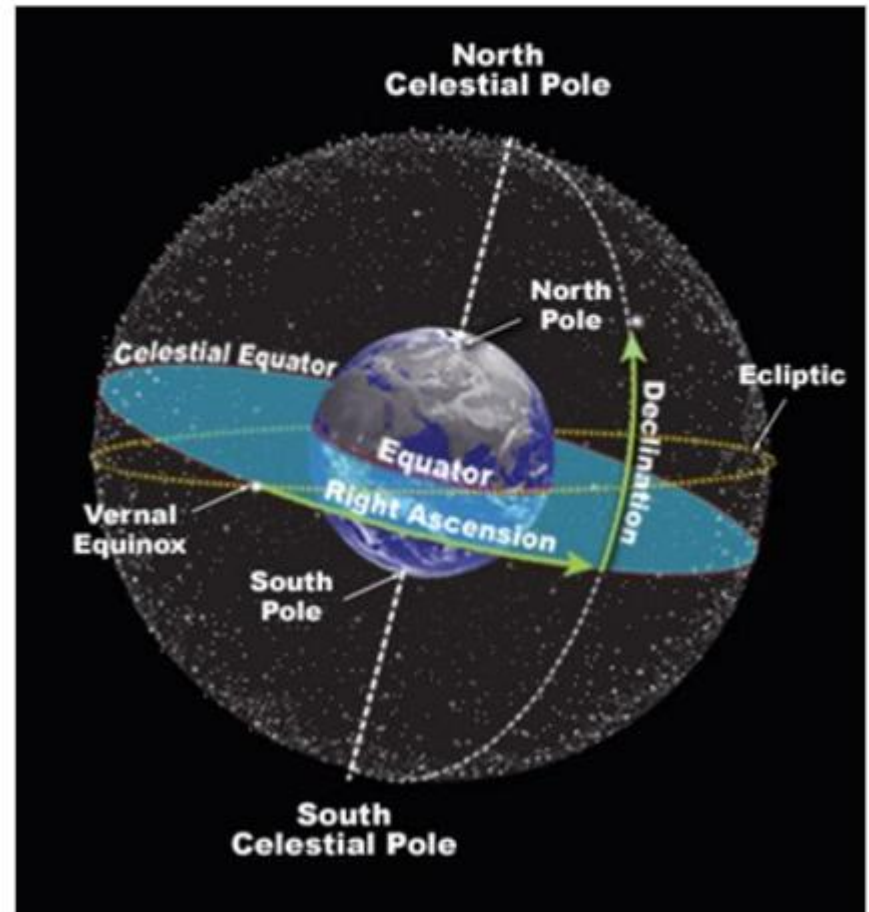
Motion of equinox and solstices

- The [solstices](#) and [equinoxes](#) signal the changing of the seasons on Earth



[Some details about our sky(1)]

- All the stars are stationary w.r.t. each other (Approx).
- Observing from earth, they create a virtual reference spherical map.
- Sun's path in the celestial sphere is called Ecliptic.



Celestial Sphere

Sidereal day:

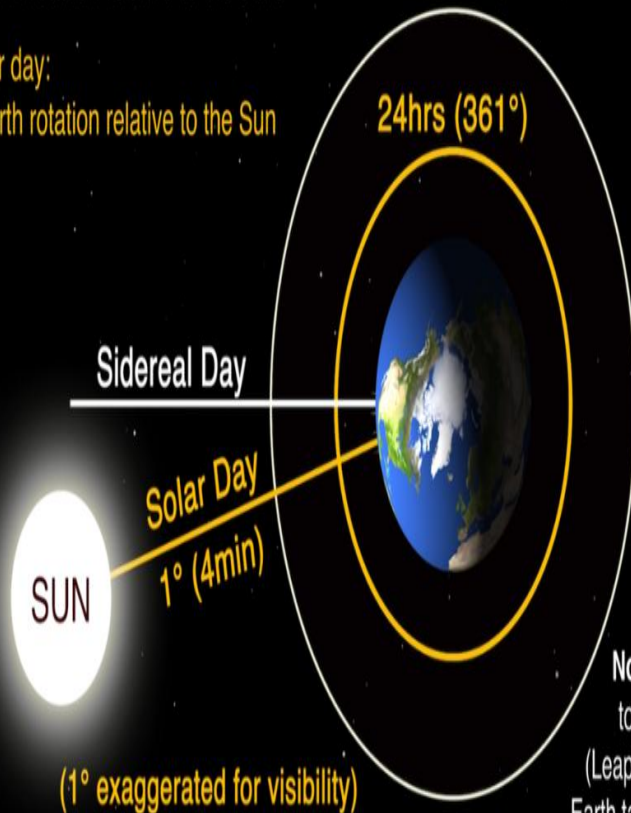
1 Earth rotation relative to the stars

23hrs 56min (360°)

Solar day:

1 Earth rotation relative to the Sun

24hrs (361°)



Earth Rotation
 361°

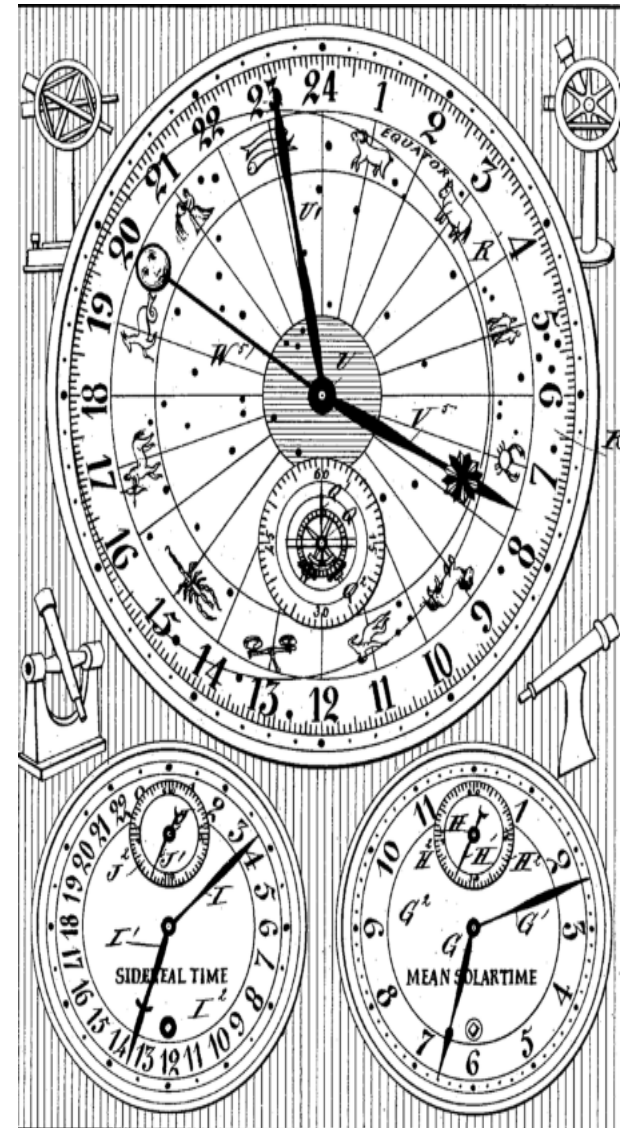
Earth rotates 366 times
per 'normal' 365-day year
relative to the stars

Note: this has nothing
to do with leap years!
(Leap years needed because
Earth takes 365.24 days to orbit)

Sidereal Day = 23hr 56min 4sec

Solar Day = 24hrs

James O'Donoghue (@physicsJ)



Indian calender system

Why we need a calander ?

We need food

|

Grow crops

|

Predict Weather

|

Wheather synchronous calender

[Units to represent Calendar!]

Let's see what natural time periods they had.

- Sun's position w.r.t. stars. (365.256 363 051 days)
- Equinox (365.242 189 67 days)
- Moon's position w.r.t. stars. (27.321 582 days)
- Moon's phases. (29.530 588 days)
- Solar day(1 day)

Problem:

Time units can only have **integer ratio** but above time periods are real numbers ☹

[Day]

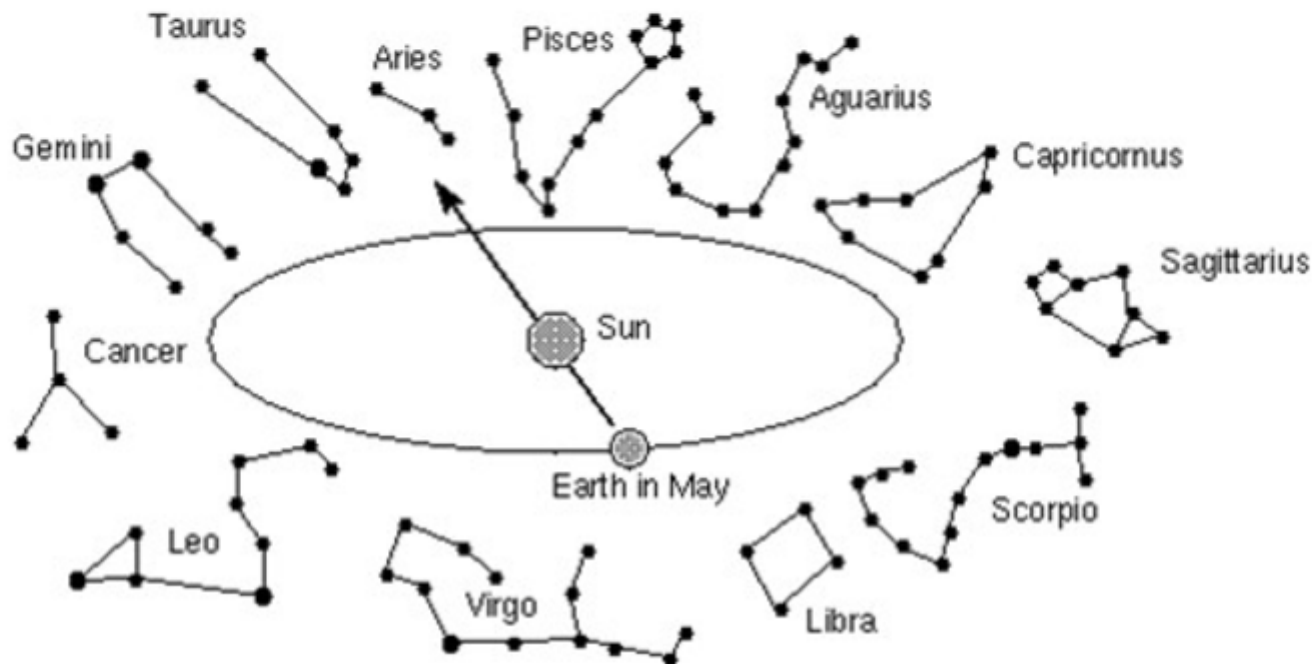
- Sunrise to sunrise
- Day has a name according to moon's phase at the time of sunrise.



- Two weeks in a month of 14/15 days each:
 - Bright week
 - Dark week
- A month = 29/30 days

[Some details about our sky(2)]

- Sun moves with respect to celestial sphere in a circle.
- Indians/Greeks divided this circle in 12 parts.
- These parts are known as Sun signs/Zodiacs.



Elements of Indian calendar systems.

- India follows four main types of calendars:
- Vikram Samvat (Hindu lunar calendar)
- Saka Samvat (Hindu solar calendar)
- Hijri Calendar (Islamic lunar calendar)
- Gregorian Calendar (Scientific solar calendar)
- The Hindu calendar, used since about 1000 BCE, is based on 12 lunar months.
- Discrepancy between the lunar year (~354 days) and solar year (~365 days) is adjusted by intercalation (extra months or days).
- The Hindu calendar has roots in the Vedic era.
- Vedic texts mention 27 nakshatras, constellations along the ecliptic, each representing roughly one day of the Moon's path.



Vikram Samvat

Vikram Samvat is the most well-known Era used currently in traditional India

- Many historians believe that Chandragupta, the king of Ujjain who took the title of Vikramaditya, started this lunisolar era in 57 BCE when he won over the Saka rulers
- It is a lunisolar calendar as per Hindu rituals from the past
- It starts around April 15th in Nepal, marking the beginning of solar's New Year
- There are 12 months in a year and 354 days in total

But still, some people believe that this calendar was created by the Ganarajya and later on, it was named after Vikramaditya

- This calendar is lunar, based on the calendar of Hindus of ancient times.

Saka Samvat

- The Sakar rulers started the Saka Samvat calendar to commemorate their victory over kushana rulers
- Many historians believe that it was founded by a king of the Satavahana dynasty named Shalivahana
- It began in 78 CE
- It marks the beginning of the Saka Samvat. Later on, in 1957, this calendar was introduced as the **Indian National Calendar**
- The Saka calendar is also referred to as the Shalivahana Saka era or Mahasakkarat Era
- In the Saka calendar, every year consists of 365 days
- It also contains a constant number of days in each month
- This calendar is used in India and used in other countries like Indonesia, Nepal, the Philippines, etc

Gregorian Calendar

- The Gregorian Calendar was introduced in October 1582
- It is based on the birthday of Jesus Christ- the founder of the Christianity religion
- It replaced the Julian Calendar due to its error in calculating the leap year
- It is named after Pope Gregory XIII
- It is the most widely accepted calendar used around the world
- It is a solar year, and it begins on 1st January and lasts for 365 days, 5 hours, 48 minutes, and 46 seconds
- The Gregorian Calendar is normally used as the Civil calendar

A hindu calender is also called Panchangam meaning “five limb”

.

The five limbs are lunar day (tithi), half day(karana), constellation (Nakshatra), angles of sun and moon (yoga), and solar weekday (vara).

Details in the next slide

The Indian calendar system, also known as the Panchang, includes several elements, including: 

- **Days of the week:** The seven-day week was adopted from Hellenistic culture. 
- **Lunar days:** The lunar day, or tithi, is the time it takes for the moon to move 12° relative to the Earth-Sun axis. 
- **Constellations:** The nakshatra, or lunar mansion, is a portion of the moon's orbit around the Earth. Each nakshatra is about one day long. 
- **Yoga:** The yoga is a portion of the combined longitudes of the moon and the sun. Each yoga is associated with certain human qualities, deities, or other mythological figures. 
- **Karaṇa:** The karaṇa is half a lunar day. Each karaṇa is associated with certain qualities. 

- **Seasons:** The year is divided into six seasons, each consisting of two months. [🔗](#)
- **Months:** The Hindu calendar months were altered in the 4th century CE. [🔗](#)
- **Solar and lunar elements:** The Hindu calendar is based on a geocentric model of the Solar System. It includes both solar and lunar elements, as well as synodic, sidereal, and tropical elements. [🔗](#)
- **Pakṣa:** The pakṣa is the time it takes for the moon to move from a new moon to a full moon, or vice versa. The waxing phase is called the śukla pakṣa and the waning phase is called the kṛṣṇa pakṣa. [🔗](#)
- **Cāndramāsa:** The cāndramāsa is the time it takes for the moon to move from one new moon to the next, or from one full moon to the next. [🔗](#)

- The year was divided into six seasons, each consisting of two months; Vedic sources had one set of month names, later revised by the 4th century CE.
- The six seasons guide the ritual calendar, though regional climates may differ.
- The Sanskrit poem Ritusamhara by Kalidasa (c. 5th century CE) describes all six seasons, blending nature with romantic imagery.
- Each month is divided into two halves:
 - Dark half: Krishnapaksha (full moon to new moon)
 - Bright half: Shuklapaksha (new moon to full moon)
- A month has roughly 30 lunar days (tithis), each lasting 21.5–26 hours, forming the basis of the panchangam system.
- The Vikram Samvat calendar year has 354 days.
- In a five-year cycle, the 3rd and 5th years have an extra (13th) month.

Six seasons of the Hindu calendar year

Sanskrit	English	<u>Gregorian calendar</u> equivalent months
<i>vasanta</i>	spring	mid-March until mid-May
<i>grishma</i>	summer, hot	mid-May until mid-July
<i>varsha</i>	rains	mid-July until mid-September
<i>sharada</i>	autumn	mid-September until mid-November
<i>hemanta</i>	winter	mid-November until mid-January
<i>shishira</i>	dews, cool	mid-January until mid-March

Hindu calendar months

name	Gregorian equivalent
Chaitra	March–April
Vaishakha (Vaisakh)	April–May
Jyeshtha (Jyaishtha)	May–June
Ashadha (Ashadh)	June–July
Shravana (Shravan)	July–August
Bhadrapada (Bhadra)	August–September
Ashvina (Ashvin)	September–October
Karttika (Kartik)	October–November
Margashirsha	November–December
Pausha (Paush)	December–January
Magha (Magh)	January–February
Phalguna (Phalgun)	February–March

Sanskrit days of the week

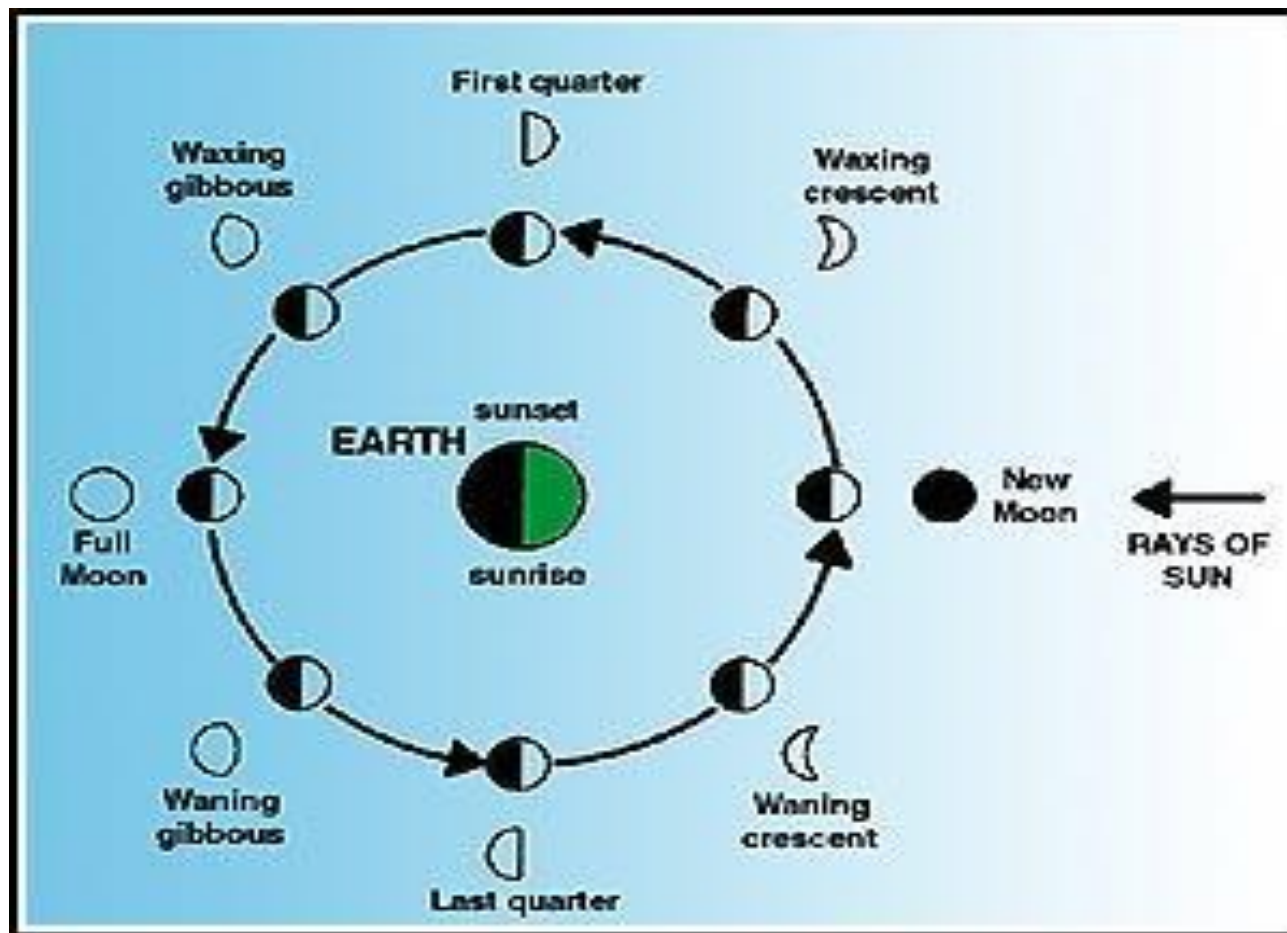
day	associated celestial body	English equivalent
Ravivasara	Sun (Surya)	Sunday
Somavasara	Moon	Monday
Mangalavasara	Mars	Tuesday
Budhavasara	Mercury	Wednesday
Guruvasara	Jupiter	Thursday
Shukravasara	Venus	Friday
Shanivasara	Saturn	Saturday

Zodiac signs (*rashis*) in the Hindu calendar

<i>rashi</i>	symbol	Western equivalent
<i>mesha</i>	ram	<u>Aries</u>
<i>vrishabha</i>	bull	<u>Taurus</u>
<i>mithuna</i>	pair	<u>Gemini</u>
<i>karkata</i>	crab	<u>Cancer</u>
<i>simha</i>	lion	<u>Leo</u>
<i>kanya</i>	maiden	<u>Virgo</u>
<i>tula</i>	scale	<u>Libra</u>
<i>vrishchika</i>	scorpion	<u>Scorpius</u>
<i>dhanus</i>	bow	<u>Sagittarius</u>
<i>makara</i>	crocodile	<u>Capricorn</u>
<i>kumbha</i>	water jar	<u>Aquarius</u>
<i>mina</i>	fish	<u>Pisces</u>

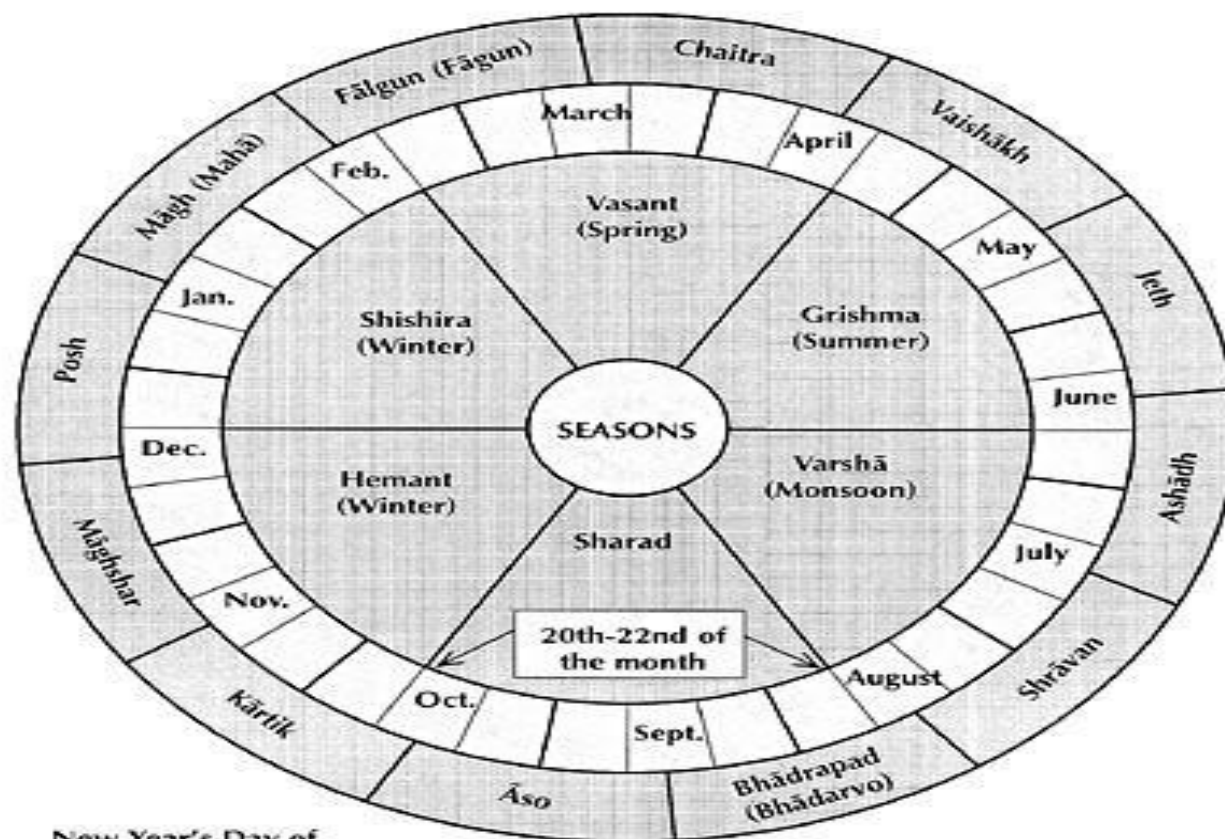
Ugadi	1st of Chaitra (bright half)
Rama Navami	9th of Chaitra (bright half)
Rathayatra	2nd of Ashadha (bright half)
Raksha Bandhan	full moon of Shravana
Janmashtami	<i>purnimanta</i> : 8th of Bhadrapada (dark half); <i>amavasyanta</i> : 8th of Shravana (dark half)
Ganesh Chaturthi	4th of Bhadrapada (bright half)
Navratri	1st to 9th of Ashvina (bright half)
Durga Puja	1st to 10th day of Ashvina (bright half)
Dussehra , Vijayadashami	10th of Ashvina (bright half)
Diwali	<i>purnimanta</i> : new moon day of Karttika; <i>amavasyanta</i> : new moon of Ashvina
Maha Shivaratri	<i>amavasyanta</i> : 13th Magha (dark half); <i>purnimanta</i> , 13th Phalguna (dark half)
Holi	full moon of Phalguna

- * Most major Hindu festivals are based on the Hindu lunar months, so their dates shift annually in the Gregorian calendar.
- * Some holidays follow the solar calendar, based on zodiac signs.
- * Makar Sankranti and Tamil Pongal mark the Sun's entry into Capricorn.
- * Some regional New Year celebrations follow the Sun entering Aries (Mesha Sankranti), including Vishu (Kerala), Puthandu (Tamil Nadu), Pohela Boishakh (Bengal), Bihu (Assam), and Vaisakhi (Punjab).
- * Otherwise, festival dates are primarily determined by the lunar calendar.

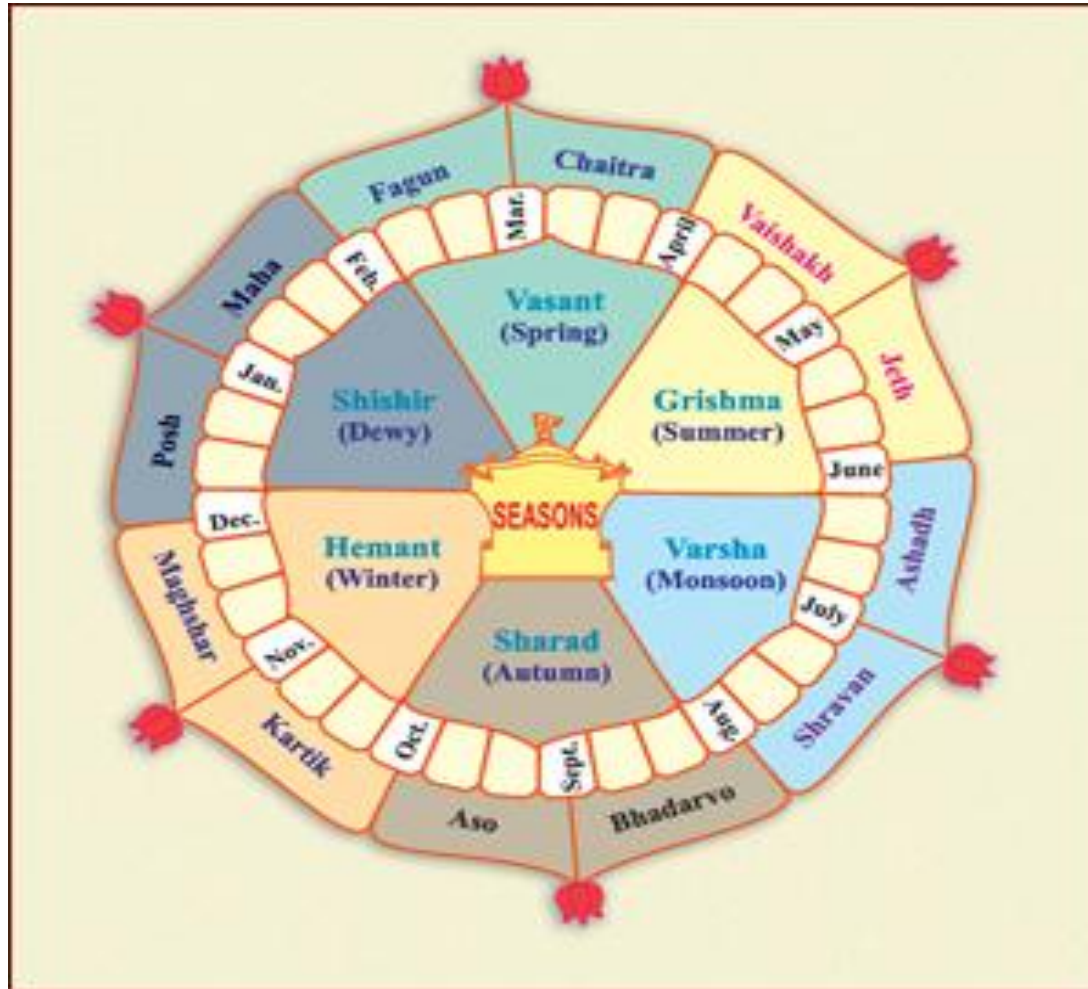


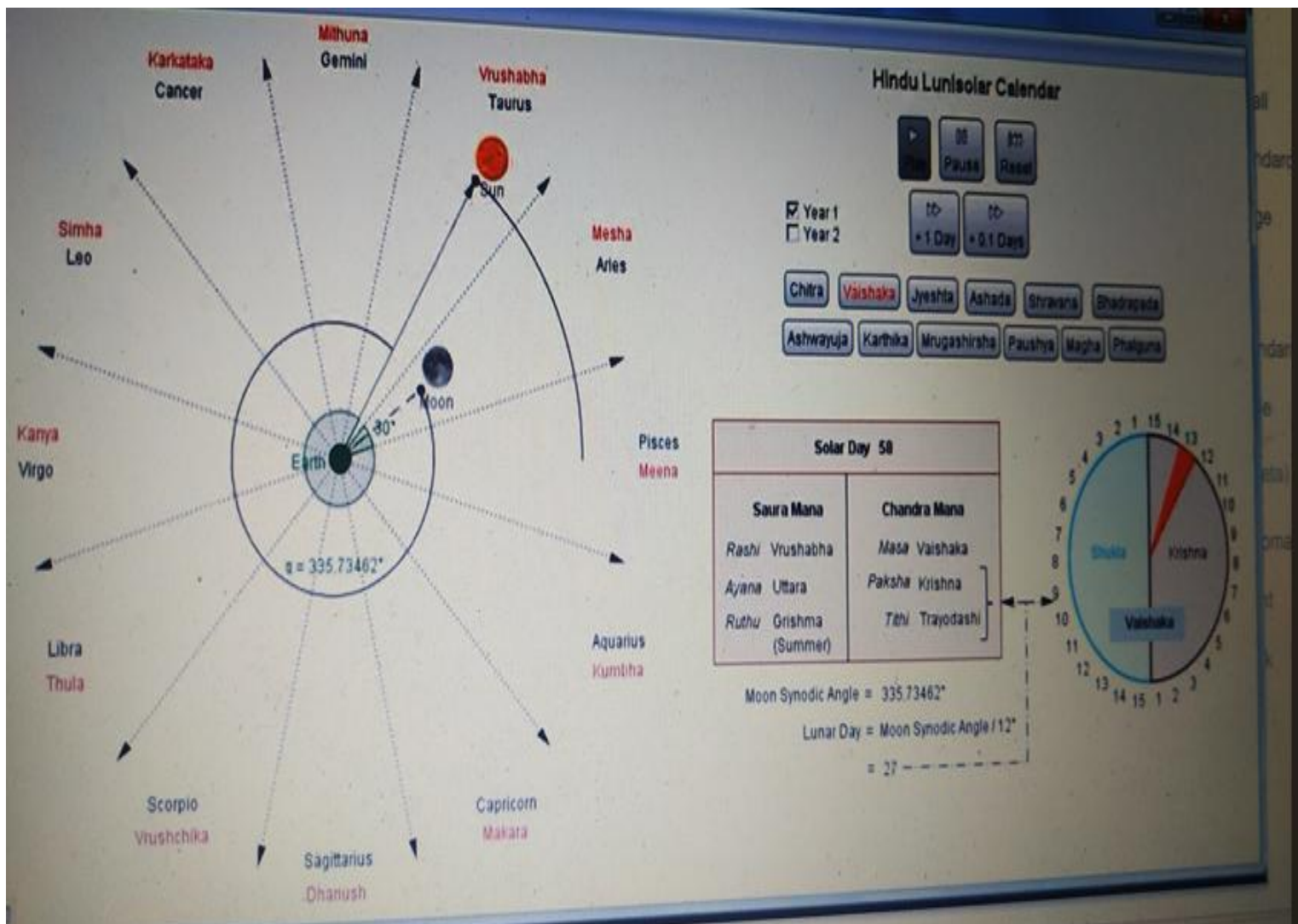
The Hindu Calendar & Seasons

(and how it relates to the calendar we use today)

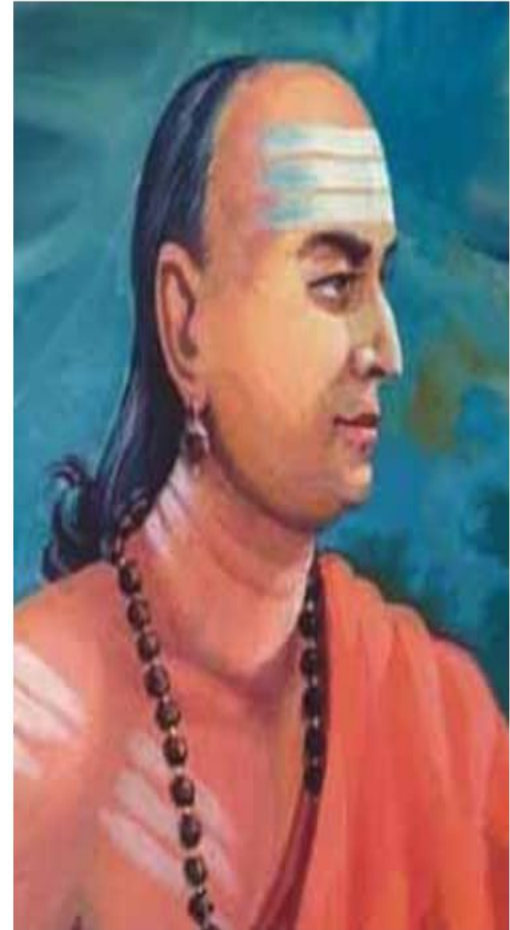


New Year's Day of
Vikram Samvat
(Kārtik Sud 1)





Basic ideas of the planetary model of Aryabhatta and its revision by Nilkantha



Basic ideas of the planetary model Aryabhata and its revision by Nilkantha

- Indian astronomy flourished in the 5th–6th century with Aryabhata.
- Aryabhata's Aryabhatiya has four sections covering units of time, planetary positions, and the cause of day and night.
- The Aryabhatiya introduced a heliocentric-like model with Earth spinning on its axis, predating Copernicus by over a millennium.
- Aryabhata accurately calculated π , the length of the solar year, and Earth's circumference.
- Native Indian astronomical tradition continued through the medieval period into the 16th–17th century, notably in the Kerala school.

- * Aryabhata proposed that the Earth rotates on its axis, explaining the apparent movement of stars and the sidereal day.
- * The Aryabhatiya does not explicitly describe Earth's heliocentric revolution; the solar year is attributed to the Sun's motion.
- * There is no direct phrase in the text clearly stating Earth's motion through space beyond axial rotation.
- * Numerical data in Aryabhatiya concerning Earth relate to the number of rotations in a yuga or the rate of rotation.
- * Verses such as Gola 9 (analogy with a moving boat) and Gitika 3 (listing revolutions of celestial bodies) suggest Aryabhata envisioned some motion of Earth in space, though indirectly.
- * His theory is asserted confidently, but the text provides no explicit use of Earth's rotation to simplify calculations or explain phenomena beyond the movement of stars.

- Aryabhata believe that the earth rotates.
- Aryabhata had explicitly stated (Gola 5 and 6) that the Earth is spherical.
- His commentators Lalla and Nilakatha explained why **the spherical Earth appeared as a plane**. Nilakantha (1500 CE) developed a heliocentric model for the planets - a significant achievement **decades before Copernicus**. He said: "Due to the large area, one does not feel that it is spherical because we see only a small part of the Earth and also because of the ups and downs on the surface of the Earth."

conclusion

- Indian astronomical knowledge spread to neighboring regions like Persia and Greece, influencing astronomers such as Al-Biruni and Ptolemy.
- Indian mathematical and observational techniques shaped astronomy in other cultures.
- Ancient Indian astronomy showcases the civilization's intellectual prowess and quest to understand the heavens.
- The discoveries, instruments, and insights of early Indian astronomers continue to inform and inspire modern astronomy.
- Their legacy reminds us of India's enduring contribution to scientific knowledge and our understanding of the cosmos.

Unit 5

Indian Health Sciences

Vedic foundations of Ayurveda. Basic concepts of Ayurveda. The three Gunas and three Doshas, Pancha-mahabhuta and Sapta-dhatu. Six Rasas and their relation to Doshas. Ayurvedic view of the cause of diseases. Dinacharya or daily regimen for the maintenance of good health. Current revival of Ayurveda and Yoga.

1. Vedic Foundations of Ayurveda:

1.1 Introduction:

Ayurveda, the ancient Indian science of life and health, is not merely a system of medicine—it is a philosophy, a way of living in harmony with nature and oneself. Its roots trace back thousands of years to the Vedic period of Indian civilization, where the earliest whispers of medical thought were woven into hymns, rituals, and cosmic visions. In the vast tapestry of the Vedas, the seeds of Ayurveda were sown—not just as practical knowledge of herbs and healing, but as sacred wisdom passed from the divine to the human realm.

To understand Ayurveda's true essence, one must first look to the Vedas, the oldest scriptures of India. Among the four Vedas—Rigveda, Yajurveda, Samaveda, and Atharvaveda—it is the Atharvaveda that holds the most direct references to health, disease, and healing. Here, Ayurveda begins its story.

1.2 Ayurveda in the Vedas:

The Rigveda, considered the oldest of the Vedas, contains hymns praising the healing powers of herbs, waters, and divine forces. It refers to certain diseases and their cures in metaphorical language, suggesting that health was seen as a divine gift and disease as a disruption of cosmic harmony.

The Atharvaveda, however, is where we find Ayurveda's true Vedic foundations. Composed between 1200 BCE and 1000 BCE, this text contains over a hundred hymns related to health, healing, herbal treatments, and even surgeries. Disease is portrayed as the result of both internal imbalance and external malevolent forces—spirits, curses, or breaches in ritual conduct.

Healing was holistic:

- **Herbs** were used for physical ailments.
- **Mantras and rituals** were employed to purify the mind and spirit.
- **Moral conduct** and self-discipline were valued as essential to well-being.

Here, we see the early formulation of Ayurveda's principle: **a balanced life—body, mind, and spirit—is the foundation of health.**

1.3 The Divine Origins of Ayurveda:

Ayurveda, according to tradition, is not of human origin. It is said to be *apauruṣeya*—not created by man, but revealed by the divine. This is consistent with how the Vedas themselves are regarded: eternal truths perceived by enlightened sages.

In the mythological account found in the *Charaka Samhita*, the knowledge of Ayurveda was first conceptualized by *Lord Brahmā*, the creator of the universe. He shared this knowledge with *Dakṣa Prajāpati*, who taught it to the divine twins, the *Ashvinī Kumāras*—physicians to the gods. From there, it passed to *Indra*, the king of the gods.

When humanity began suffering from increasing disease and suffering during a time of great strife, a sage named *Bhāradvāja* ascended to the heavens to request this knowledge from Indra. The god consented and imparted Ayurveda to Bhāradvāja, who then brought it to Earth. Bhāradvāja taught the science to *Ātreya Punarvasu*, who in turn trained six disciples—including the famed *Agniveśa*, whose work forms the basis of the Charaka Samhita.

Thus, Ayurveda is a *divine gift to humanity*, designed not just to cure disease, but to preserve harmony in human life and its relationship with the cosmos.

1.4 Philosophical Foundations:

While its roots lie in the spiritual and ritualistic ethos of the Vedas, Ayurveda's theoretical structure matured through Vedic philosophy, especially the *Sāṅkhya* and *Vaiśeṣika* schools of thought.

Sāṅkhya Philosophy

Sāṅkhya, attributed to the sage Kapila, explains creation through the interaction of *Purusha* (pure consciousness) and *Prakriti* (primordial nature). Prakriti evolves into the material world through the process of manifestation—from subtle elements (*tanmātras*) to gross elements (*mahābhūtas*).

This model forms the backbone of Ayurvedic metaphysics:

- The human being is a microcosm of the universe.
- The body and mind arise from the *five great elements (pañca-mahābhūtas)*: Pṛthvī or Kṣiti (earth), Ap (water), Tejas (fire), Marut or Vāyu (air) and Vyom or Ākāśa (ether or sky).
- Health depends on the *balance* of these elements within.

Vaiśeṣika & Nyāya Systems

These philosophies contributed the *logic, categories of substances*, and methods of inference. Ayurveda borrows from Vaiśeṣika's categorization of reality into six categories (*padārthas*): *dravya* (substance), *guṇa* (quality), *karma* (action), *sāmānya* (generality), *viśeṣa* (particularity), and *samavāya* (inherence).

Thus, Ayurveda is both a *science of the body* and a *science of knowledge*. It respects empirical observation but is rooted in *cosmic principles*.

1.5 Ayurveda Becomes a Medical System:

As centuries passed, Ayurvedic knowledge was compiled, refined, and expanded into structured treatises. These were no longer only spiritual discourses but *systematic medical texts* covering anatomy, pathology, diagnosis, treatment, pharmacology, surgery, and preventive care.

This transformation gave rise to the *classical triad* of Ayurvedic literature—known as the *Bṛhat Trayī (Great Triad)*:

1. **Charaka Saṃhitā** – Focused on internal medicine (*kāyacikitsā*) and general principles of Ayurveda.
2. **Suśruta Saṃhitā** – Known for detailed surgical methods and anatomy.
3. **Aṣṭāṅga Hṛdaya** – A poetic summary of the two earlier works, encompassing all eight branches of Ayurveda (*Aṣṭāṅga Ayurveda*).

These texts were written not as religious scriptures but as *clinical manuals*, based on observation, dialogue, and experience. Their influence endured for centuries and still forms the core of Ayurvedic education today.

1.6 From Cosmic Order to Human Health:

What makes Ayurveda unique is its understanding of health not merely as the absence of disease, but as a *state of dynamic equilibrium*—between the *body's inner constitution (prakriti)* and the *external world (kāla, deśa, and karma)*.

Health is defined in the *Suśruta Saṃhitā* as follows:

*"Samadoṣaḥ samāgniśca samadhātu malakriyāḥ
Prasannātmendriyamanāḥ svastha ityabhidhīyate"*

– Suśruta Saṃhitā, Sūtrasthāna 15.41

“One who has balanced doshas, balanced digestive fire (agni), properly formed dhatus and malas, and a contented soul, mind, and senses, is truly healthy.”

This holistic definition reflects Ayurveda’s *Vedic foundation*—a worldview where *disease is disharmony*, and health is *harmony with the rhythms of the universe*.

The Vedic foundations of Ayurveda give it a timeless, sacred dimension. It is not only a science of medicine but a philosophy of life—an integrated system where physical health, mental clarity, and spiritual well-being are inseparable.

Born in the age of hymns and rituals, shaped by the rationality of philosophical schools, and matured through centuries of clinical observation, Ayurveda remains a profound expression of India's ancient wisdom. It invites each individual to live in balance—with nature, with time, and with oneself.

2. Basic Concepts of Ayurveda:

2.1 Introduction:

At the heart of Ayurveda lies a profound vision of life—*not as a collection of isolated parts*, but as a *dynamic, interconnected system*. Unlike modern biomedical systems that often reduce health to measurable symptoms, Ayurveda embraces a *holistic worldview*. It sees the human being as a *microcosm of the universe*, formed from the same elemental building blocks and governed by the same natural laws.

To truly understand Ayurveda, one must first grasp its basic conceptual framework: the *Five Great Elements (Pañcha Mahābhūtas)*, the *Three Doshas*, the *Three Gunas*, the *Seven Dhatus*, and the *notion of balance* that underpins all Ayurvedic thought. These are not abstract ideas, but living principles—guiding diagnosis, treatment, diet, and lifestyle.

2.2 The Five Great Elements (Pañcha Mahābhūtas):

In Ayurveda, all matter in the universe—including the human body—is composed of five fundamental elements, called *Mahābhūtas*. These are:

1. **Ākāśa (Ether/Space):** Represents openness, subtlety, and sound. It forms cavities in the body such as the mouth and gastrointestinal tract.
2. **Vāyu (Air):** Represents movement and touch. It governs all motion, including breathing, circulation, and nerve impulses.
3. **Tejas or Agni (Fire):** Represents transformation, digestion, vision, and metabolism.
4. **Āpas (Water):** Represents cohesion, liquidity, and taste. Found in plasma, mucus, and secretions.
5. **Prthvī (Earth):** Represents solidity, structure, and smell. It is the foundation of bones, muscles, and organs.

These elements are *not separate*, but always exist in dynamic interaction. Every object, living being, and experience is a unique combination of these five elements.

Ayurveda teaches: यत् पिण्डे तत् ब्रह्माण्डे (Yat piṇḍe tat brahmāṇḍe)

– “As in the microcosm, so in the macrocosm.”

The same elements that make up the universe also make up the body.

2.3 The Three Doshas: Vāta, Pitta, and Kapha:

The five elements combine to form three functional forces or *bio-energies* in the body, known as *Doshas*:

Dosha	Elements	Primary Function
Vāta	Ether + Air	Movement, circulation, communication

Pitta	Fire + Water	Digestion, metabolism, transformation
Kapha	Earth + Water	Structure, lubrication, stability

- Vāta Dosha
 - Governs all movement: breathing, heartbeat, nerve impulses, muscle contraction.
 - Qualities: dry, light, cold, rough, mobile, subtle.
 - Imbalance can cause anxiety, constipation, dryness, joint pain.
- Pitta Dosha
 - Controls digestion, body temperature, and vision.
 - Qualities: hot, sharp, oily, liquid, sour.
 - Imbalance leads to acidity, anger, rashes, inflammation.
- Kapha Dosha
 - Provides structure, immunity, and emotional calm.
 - Qualities: heavy, slow, soft, cold, oily, stable.
 - Imbalance causes weight gain, lethargy, congestion, attachment.

Every individual is born with a unique *constitution (Prakṛti)*—a specific balance of these three doshas. Maintaining this balance is key to health. When a dosha increases or decreases beyond its natural limits, *disease (vikṛti)* arises.

2.4 The Three Gunas: Sattva, Rajas, and Tamas:

While the doshas govern the *body*, the *gunas* govern the *mind*.

Guna	Nature	Mental Effects
Sattva	Purity, harmony	Clarity, peace, wisdom
Rajas	Activity, passion	Desire, ambition, restlessness
Tamas	Inertia, darkness	Ignorance, confusion, lethargy

- ✓ **Sattva** is the ideal mental state: balanced, content, and spiritually inclined.
- ✓ **Rajas** leads to ambition and action, but in excess causes anxiety and aggression.
- ✓ **Tamas** is necessary for rest, but in excess leads to depression and ignorance.

Ayurveda teaches that **mental health depends on cultivating sattva** through ethical conduct, clean food, meditation, and self-awareness.

2.5 The Seven Dhatus (Tissues):

The physical body is built from *seven fundamental tissues*, known as *Dhatus*. Each dhatu nourishes the next in a *sequential chain of transformation*.

Dhatu	Function
Rasa (Plasma)	Nourishment, fluid transport
Rakta (Blood)	Oxygenation, vitality
Māṃsa (Muscle)	Strength, movement
Meda (Fat)	Lubrication, insulation
Asthī (Bone)	Support, structure
Majjā (Marrow)	Nervous system, regeneration
Śukra / Artava (Reproductive Tissue)	Fertility, creativity

If digestion and metabolism (Agni) are strong, dhatus are properly nourished. Poor nutrition or weak digestion disrupts this chain and leads to disease.

2.6 Agni: The Digestive Fire:

No concept is more central to Ayurvedic health than *Agni*, the *digestive fire*. It is the force of transformation, converting food into energy and tissues.

There are 13 types of Agni, with *Jatharagni* (digestive fire in the stomach) being the most important. The Functional states of Jaṭharāgni (strength-wise):

- ✓ समाग्नि (Samāgni) – balanced, regular digestion
- ✓ विषमाग्नि (Viśamāgni) – irregular (follows Vāta dominance)
- ✓ तीक्ष्णाग्नि (Tikṣṇāgni) – excessively fast digestion (Pitta dominance)
- ✓ मन्दाग्नि (Mandāgni) – slow digestion (Kapha dominance)

Sl.	Type of Agni (Charaka)	Meaning
1	जठराग्नि • <i>Jaṭharāgni</i>	Central digestive fire
2	भौमाग्नि • <i>Bhāvāgni</i>	Earth-element fire (Parthivā)
3	आप्याग्नि • <i>Āpyāgni</i>	Water-element fire (Āpya)
4	आग्नेयाग्नि • <i>Āgneyāgni</i>	Fire-element fire (Tejasa)
5	वायव्याग्नि • <i>Vāyavyāgni</i>	Air-element fire (Vāyu)
6	नाभसाग्नि • <i>Nābhasāgni</i>	Ether-element fire (Ākāśa)
7	रसाग्नि • <i>Rasāgni</i>	Plasma/tissue fire (Rasa)
8	रक्ताग्नि • <i>Raktāgni</i>	Blood tissue fire (Rakta)
9	मांसाग्नि • <i>Māṃsāgni</i>	Muscle tissue fire (Mamsa)
10	मेदाग्नि • <i>Medāgni</i>	Fat/adipose tissue fire (Meda)
11	अस्थिग्नि • <i>Asthīagni</i>	Bone tissue fire (Asthi)
12	मज्जाग्नि • <i>Majjāgni</i>	Bone-marrow/nervous tissue fire (Majjā)
13	शुक्राग्नि • <i>Śukrāgni</i>	Reproductive tissue fire (Śukra)

Balanced Agni means: Good appetite, Clear thoughts, Strong immunity, Regular elimination
 Disturbed Agni leads to: Toxins (Ama), Indigestion, Sluggishness, Inflammation
 Maintaining strong Agni is the *cornerstone of Ayurvedic health*.

2.7 Mala: Waste Products:

Just as the body produces useful tissues, it also produces *wastes* that must be properly eliminated. These are called Mala:

Purīṣa – feces, **Mūtra** – urine, **Sveda** – sweat

Proper elimination of malas indicates that *doshas, dhatus, and agni are functioning well*.

2.8 Prakṛti and Vikṛti: Constitution and Imbalance:

Every person is born with a unique *Prakṛti*—their natural combination of doshas and gunas. It is *determined at conception* and remains constant throughout life.

Vikṛti refers to the current state of imbalance caused by:

Diet and lifestyle errors, Seasonal changes, Emotional stress, Improper routines

Ayurveda seeks to *bring the body back to its prakṛti*, its original state of balance. This is the true goal of treatment—not just to remove symptoms, but to restore natural harmony.

2.9 The Path to Health:

Ayurveda defines health not only as the absence of disease, but as a state of total harmony.

“**Swasthya** is a state where doshas, agni, dhatus, and malas are balanced, and the soul, senses, and mind are in a state of bliss.”

Health is a *dynamic balance*, requiring:

Proper digestion, Balanced routines, Clean thoughts, Harmonious relationships, Seasonal adaptation, Inner awareness.

In this way, Ayurveda is not just a medical science, but a *science of life (Ayur–Veda)*—a timeless guide to living with intelligence and grace.

The basic concepts of Ayurveda reveal a system that is deeply philosophical, but also intensely practical. It is a science that honors individuality, promotes self-awareness, and seeks to understand the *why* behind disease—not just the *what*.

Understanding doshas, gunas, dhatus, and agni offers more than knowledge. It provides a *blueprint for living*—where the human body, mind, and soul are seen not as separate entities, but as interconnected expressions of nature's intelligence.

Ayurveda invites us to live in balance—with our constitution, our environment, and the eternal rhythms of life.

3. The Ayurvedic View of the Cause of Diseases:

3.1 Introduction:

In modern medicine, disease is often attributed to microbes, genetic defects, or biochemical imbalances. Ayurveda, while not denying physical causes, looks *deeper and broader*. Disease, according to Ayurveda, is not an isolated event. It is the *final symptom of a deeper disturbance* in the body's balance—between elements, energies, mind, senses, environment, and spirit.

Ayurveda is a science of *balance*. It believes that health is the natural state of the body, and disease arises only when that balance is disturbed. Therefore, to prevent or cure disease, one must understand *why* imbalance occurs—not just how it manifests.

3.2 Health: A Balanced State:

Before understanding disease, we must understand *health* in Ayurveda.

Health (swasthya) is defined in the *Suśruta Saṁhitā* as:

समदोषः समाग्निश्च समधातु मलःक्रियाः ।
प्रसन्नात्मेन्द्रियमनाः स्वस्थ इत्यभिधीयते ॥

“Samadoṣaḥ samāgniśca samadhātu malakriyāḥ,
Prasannātmendriyamanāḥ svastha ityabhidhīyate.”

(Suśruta Saṁhitā, Sūtrasthāna 15.41)

“One whose doshas, digestive fire (agni), tissues (dhatus), and waste products (malas) are in balance, and whose soul, senses, and mind are in a pleasant state, is considered healthy.”

Therefore, the cause of disease is anything that *disturbs* this equilibrium.

3.3 Root Causes of Disease in Ayurveda:

1. Pragyāparādha – Intellectual Error (Misuse of the Intellect)

This is considered the *primary cause of all disease*.

- Pragyā = wisdom, discernment
- Aparādha = offense, mistake

Pragyāparādha means the *failure of the intellect to make proper decisions*. It causes people to go against nature, to adopt unhealthy habits knowingly, or to ignore signals from the body.

Examples include:

- Eating unhealthy food despite knowing it's harmful
- Suppressing natural urges (like hunger or sleep)
- Letting emotions overpower reason
- Pursuing stressful lifestyles

Ayurveda teaches that when the intellect is misused, it leads to wrong actions, which in turn cause imbalances in the body.

2. Asātmayendriyārtha Saṁyoga – Misuse of the Senses

This refers to *improper connection between the senses and their objects*—either by overuse, underuse, or misuse.

Sense Organ	Misuse Examples
Eyes	Excess screen time, staring at bright lights
Ears	Constant loud noise or music
Tongue	Overeating, unhealthy food
Nose	Exposure to pollution or toxic smells
Skin	Excess heat, cold, or physical contact

When the senses are *overstimulated or neglected*, the mind becomes disturbed and *doshas begin to aggravate*, leading to disease.

3. Parināma or Kāla – Seasonal & Time-related Factors

Time (Kāla) is a powerful force in Ayurveda. Disease can be caused by *failure to adapt to time*:

- ✓ *Seasonal changes (ṛtu)*: Each season naturally alters the dominance of doshas. Ignoring seasonal diets or habits can cause imbalance.
- ✓ *Aging*: As we age, bodily resilience weakens. Not adjusting lifestyle with age leads to disease.
- ✓ *Biological rhythms*: Sleeping or eating at improper times disrupts internal cycles (circadian rhythms), disturbing digestion and immunity.

Ayurveda emphasizes *ṛtucaryā* (seasonal regimen) and *dinacharyā* (daily regimen) to prevent seasonal diseases.

3.4 Internal and External Causes:

Ayurveda recognizes both *internal (nija)* and *external (āgantū)* causes of disease.

- ✓ Internal Causes (*Nija Nidāna*)
These originate *within the body*, often due to:
 - Poor digestion (*agni-mandya*)
 - Accumulation of toxins (*āma*)
 - Dosha imbalance
 - Emotional factors like stress, anger, or grief
- ✓ External Causes (*Āgantū Nidāna*)
These arise from *external agents*, such as:
 - Injuries or trauma
 - Microorganisms (recognized as invisible causes)
 - Poisons or toxins
 - Environmental factors: pollution, climate

Ayurveda states that external causes only lead to disease *if internal resistance is weak*. Therefore, *strengthening inner balance* is the key to protection.

3.5 The Role of Agni and Āma:

- ✓ Agni – The Digestive Fire
When Agni is strong, the body digests food properly and nourishes all tissues.
When Agni is weak or irregular:
 - Food is improperly digested
 - Toxins accumulate
 - Doshas are disturbed
 - Disease follows
- ✓ Āma – The Toxin
Āma is *undigested metabolic waste*—sticky, heavy, foul—formed when food is not properly digested. Symptoms of āma:
 - Coated tongue

- Foul breath
- Bloating
- Lethargy
- Joint pain

Most chronic diseases in Ayurveda involve āma. Eliminating it is a key goal of Ayurvedic treatment.

3.6 Stages of Disease Development (Shat Kriyā Kāla):

Ayurveda describes the *progression of disease in six stages*, known as the *Ṣaṭ Kriyā Kāla*:

Stage	Name	Meaning
1	Sanchaya	Accumulation of dosha
2	Prakopa	Aggravation
3	Prasara	Spread to other areas
4	Sthānasamshraya	Localization in a specific tissue
5	Vyakti	Clinical manifestation of disease
6	Bheda	Complications or chronic state

A skilled Ayurvedic practitioner can detect disease even in the *first two stages*, long before symptoms appear—allowing for early correction and prevention.

3.7 Emotions and Lifestyle as Disease Triggers:

Ayurveda recognizes that the *mind and body are deeply connected*. Emotional imbalances can disrupt doshas and digestion. Examples: Fear and anxiety increase *Vāta*, Anger increases *Pitta*, Attachment and greed increase *Kapha*.

Similarly, improper lifestyle—overeating, sedentary habits, poor sleep, or irregular routines—can weaken agni and disturb doshas.

Ayurveda emphasizes *discipline (niyama)*, *self-awareness*, and *emotional regulation* as keys to preventing disease.

3.8 Disease as Loss of Harmony:

In Ayurveda, disease is not seen as an enemy, but as a *signal*—a warning from the body that something is out of alignment. Its primary causes include:

- Mistakes in judgment (Pragyāparādha)
- Misuse of the senses
- Failure to adapt to time and seasons
- Weak digestion and toxin buildup (āma)
- Emotional and lifestyle disturbances

The path to healing is not only through medicine, but through:

Restoring agni, Removing āma, Rebalancing doshas, Correcting mental and sensory habits, Living in rhythm with nature.

4. Dinacharya – The Ayurvedic Daily Regimen for Health:

4.1 Introduction:

In the hustle of modern life, health often becomes a reactive pursuit—we seek treatment after we fall ill. Ayurveda, in contrast, places great emphasis on *preventive care*, advocating that health is maintained not through medicine, but through *rhythm*—living in tune with nature’s cycles. This philosophy is beautifully embodied in *Dinacharya*, the Ayurvedic concept of a *daily regimen*.

The term *Dinacharya* comes from two Sanskrit words:

“*Dina*” = day and “*Charya*” = conduct or routine

Dinacharya is the *daily discipline* of aligning one’s bodily functions and mental activity with the natural flow of time. According to Ayurveda, such alignment strengthens digestion (*agni*), maintains doshik balance, prevents disease, and brings clarity to the mind.

4.2 The Science Behind Dinacharya:

Ayurveda teaches that *time (kāla)* is not neutral—it governs the movement of the *three doshas* throughout the day. Just as doshas dominate in seasons and stages of life, they also influence specific *times of the day and night*.

Time of Day	Dominant Dosha	Effect
6 am – 10 am	Kapha	Slow, heavy, grounded
10 am – 2 pm	Pitta	Sharp, focused, digestive
2 pm – 6 pm	Vāta	Light, active, creative
6 pm – 10 pm	Kapha	Calm, drowsy
10 pm – 2 am	Pitta	Internal metabolism
2 am – 6 am	Vāta	Light sleep, dreaming

Dinacharya is designed to *ride the wave of these natural energies*, allowing the body and mind to operate at their optimal state.

4.3 The Ayurvedic Daily Routine (Dinacharya):

Let’s now explore the ideal daily schedule as prescribed by classical Ayurvedic texts like *Ashtanga Hridaya* and *Charaka Samhita*.

1. Brahma Muhurta – Wake Before Sunrise (approx. 4:30 – 6:00 am)
 - “Brahme muhūrte uttiṣṭhet, swastho rakṣārtham āyusaḥ.”
 - “A healthy person should wake up in Brahma Muhurta to protect their life.”

This time is *dominated by Vāta*—ideal for lightness, clarity, and spiritual practice.

Waking early supports mental clarity, calmness, and longevity.

2. Evacuation (Malotsarga)
 - Attend to *natural urges*: bowel and bladder movement.
 - Suppressing these urges (*vega dharana*) is a key cause of disease in Ayurveda.
 - Regular morning elimination indicates balanced *Agni* and doshas.
3. Oral Hygiene
 - Tongue scraping (*Jihvā Nirlekhan*): Removes *āma*, stimulates organs, enhances taste.
 - Tooth brushing (*Danta Dhāvana*): Herbal sticks like neem or licorice are used.
 - Oil pulling (*Gandusha/Kavala*): Holding sesame or coconut oil in the mouth strengthens gums, removes toxins, and improves voice.
4. Nasya (Nasal Therapy)
 - Application of medicated oil or ghee into the nostrils.
 - Clears sinuses, enhances brain function, prevents respiratory disorders.
 - Best performed after cleansing and before food.
5. Abhyanga (Oil Massage)
 - Full-body massage using warm oil, especially sesame.
 - Nourishes skin and tissues, strengthens muscles, improves circulation.
 - Balances *Vāta*, calms the nervous system, delays aging.
6. Snāna (Bathing)
 - Cleanses the body, refreshes the mind, and removes excess oil.
 - Enhances circulation and improves skin tone.
 - Cold water is recommended for the body; warm water for the head is generally avoided.
7. Vyāyāma (Exercise)
 - “Ārogyam bhūtapāṁstuśca vāyuvṛddhi vināśanam.”
 - “Exercise brings strength, firmness, and lightness to the body.”

- Stimulates agni, burns āma, improves mood and immunity.
- Should be done in moderation (until light sweating). Overexertion vitiates Vāta and Pitta.
- Yoga, walking, and Surya Namaskar are commonly recommended.

8. Prāṇāyāma and Meditation

- Breathing techniques regulate *Prāṇa*, calm the mind, and balance doshas.
- Meditation enhances *Sattva*, reduces Rajas and Tamas.
- Best practiced after exercise and before breakfast.

9. Breakfast and Meals (Āhāra)

- Light, nourishing, and suitable for your *prakṛti* (constitution).
- The *main meal* should be around *midday (Pitta time)* when digestion is strongest.
- Dinner should be light and taken before sunset or by 7 pm.

10. Work and Mental Activities

- Mid-morning to early afternoon is ideal for focused tasks.
- Pitta is dominant—enhancing intellect, concentration, and metabolism.

11. Evening Wind-Down

- After sunset, avoid stimulating activities.
- Eat a light dinner, walk slowly afterward.
- Engage in relaxing, sattvic activities—reading, prayer, family time.

12. Sleep (Nidrā)

- Ideal sleep time: 10 pm
- Avoid staying awake into the Pitta hours (10 pm – 2 am), which leads to mental restlessness.
- Quality sleep maintains tissue repair, mental clarity, and doshik balance.

“Proper sleep (nidrā), balanced diet (āhāra), and control of desires (brahmacharya) are the three pillars of health.” — *Charaka Samhita*

4.4 Benefits of Dinacharya:

By following Dinacharya, one cultivates:

- ✓ Strong digestive fire (Agni)
- ✓ Balanced doshas
- ✓ Sharp mental focus
- ✓ Emotional stability
- ✓ Slower aging
- ✓ Disease prevention
- ✓ Higher sattva (mental clarity and peace)

4.5 Modern Challenges and Adaptation:

In the modern era, adhering strictly to Dinacharya can be difficult. However, even *partial adoption* brings significant benefits.

- ✓ Wake up earlier
- ✓ Scrape your tongue
- ✓ Oil your body once or twice a week
- ✓ Meditate for 5–10 minutes daily
- ✓ Eat meals on time

With consistency, these habits align your inner biology with nature’s rhythm.

4.6 Discipline for a Harmonious Life:

Dinacharya is not just a list of healthy habits—it is a *spiritual practice*, a commitment to honour the body as a temple of consciousness. Ayurveda teaches that *health is a by-product of rhythm*, and Dinacharya is the most reliable path to that rhythm.

By living each day in alignment with nature, you protect your body from disease, sharpen your mind, and open your heart to higher awareness. In a world of complexity, Dinacharya is a return to

simplicity—and to the timeless wisdom that *health is not something to be chased, but something to be lived*.

5. The Current Revival of Ayurveda and Yoga:

For millennia, *Ayurveda and Yoga* stood as pillars of Indian civilization—providing not just health systems but complete philosophies of life. These two sister sciences, deeply rooted in the *Vedas*, are as spiritual as they are scientific. However, during colonial rule and the rise of Western medicine, they were marginalized and labelled as “traditional” or “alternative.”

In recent decades, a remarkable *revival* has taken place—both in India and globally. From rural health centres to elite wellness retreats, from academic research to international legislation, Ayurveda and Yoga have re-emerged as vital contributors to modern health and well-being.

5.1 Historical Background of Decline and Preservation:

During the *colonial era*, Ayurveda and Yoga faced significant decline due to:

- The dominance of allopathic (Western) medicine in education and healthcare
- Suppression of indigenous knowledge systems
- Lack of standardization and documentation in traditional practices

Yet, these sciences *survived* through:

- Oral transmission by families of traditional healers (*vaidyas*)
- Preservation of ancient texts like *Charaka Samhita*, *Ashtanga Hridaya*, *Yoga Sutras*
- Continued use of home remedies and spiritual practices in Indian households

Revival efforts began during India's *freedom struggle*, with key leaders like *Mahatma Gandhi* advocating for indigenous health systems.

5.2 Factors Contributing to the Modern Revival:

Several factors have played a role in the 20th and 21st-century revival of Ayurveda and Yoga:

1. Scientific Validation and Research
 - Institutions like *CCRAS* (Central Council for Research in Ayurvedic Sciences) and *NIMHANS* conduct clinical studies.
 - Evidence-based benefits of yoga for stress, diabetes, hypertension, and depression are now well-documented.
 - Ayurvedic herbs like *Ashwagandha*, *Turmeric*, and *Guduchi* are studied for their pharmacological properties.
2. Global Wellness Movement
 - Rising interest in *holistic health*, *detoxification*, *mind-body medicine*.
 - Yoga is widely practiced in over 190 countries.
 - Ayurveda is promoted in wellness resorts, spas, and integrative medical clinics.
3. Government Initiatives in India
 - Creation of *Ministry of AYUSH* (Ayurveda, Yoga & Naturopathy, Unani, Siddha, Homeopathy) in 2014.
 - National AYUSH Mission promotes infrastructure and education.
 - Yoga integrated into schools, colleges, and military training.
 - *Standardization of Ayurvedic drugs* under pharmacopoeia guidelines.
4. UN and Global Recognition
 - *International Day of Yoga* (June 21) was declared by the United Nations in 2014.
 - WHO is working with India to develop a *global strategy for traditional medicine*.
 - Ayurveda is being recognized as a complementary therapy in many countries.

5.3 Revival in Education and Academia:

✓ Higher Education

- Universities across India now offer BAMS (Bachelor of Ayurvedic Medicine & Surgery) and MD (Ayurveda) degrees.
- Integration of Ayurveda into *AYUSH departments* in medical colleges.

- ✓ Digital Learning & Research
 - Online courses on Ayurveda and Yoga are available through NPTEL, Coursera, etc.
 - Research papers in reputed journals (e.g., Journal of Ayurveda and Integrative Medicine) give credibility to ancient knowledge.

5.4 Technological Integration:

The revival is also marked by integration with *modern technology*:

- **Mobile apps** for prakriti analysis, yoga tracking, dosha balance, herbal remedies.
- **Telemedicine platforms** offering Ayurvedic consultations.
- **AI tools** helping diagnose dosha imbalance and suggest personalized Ayurvedic regimens.

5.5 Globalization of Ayurveda and Yoga:

- ✓ Ayurveda Abroad:
 - Practiced in the US, Europe, Canada, Australia, and Southeast Asia.
 - Ayurvedic products and herbal supplements sold worldwide.
 - Wellness tourism to India for Ayurvedic treatment is growing.
- ✓ Yoga's Global Impact:
 - Taught in schools, offices, prisons, and therapy centres.
 - Endorsed by celebrities, athletes, and psychologists.
 - Yoga Therapy is now a recognized discipline in some parts of the world.

5.6 Challenges in the Revival Process:

Despite its growth, the revival of Ayurveda and Yoga faces several *challenges*:

Challenge	Description
Standardization	Variation in practice and interpretation across regions
Scientific Scrutiny	Need for more robust clinical trials
Commercialization	Dilution of traditional values in favor of profit
Misuse and Misbranding	False claims in herbal products and yoga therapies
Lack of Global Regulation	Inconsistent legal status in different countries

Ayurveda especially needs better *regulatory frameworks* for quality control, and *integration with public health systems*.

5.7 Future Directions:

The future of Ayurveda and Yoga depends on a balanced approach between *tradition and innovation*.

- ✓ Opportunities:
 - Ayurveda 2.0: Personalized medicine based on prakriti and genomics
 - Yoga Therapy: Integration into hospitals and rehabilitation centres
 - Global Ayurveda Centres: Like Ayurveda hospitals and wellness villages abroad
 - Curriculum Integration: Indian Knowledge System in schools and universities
 - Eco-conscious living: Use of herbal, natural, sustainable healthcare systems

India can become a *global leader in preventive healthcare* by nurturing this ancient wisdom through modern tools and ethics.

5.8 Reclaiming Our Health Heritage:

The revival of Ayurveda and Yoga is more than a trend—it is a *return to roots*, a reclaiming of ancient insights that remain profoundly relevant in our modern world. These systems remind us that health is not merely the absence of disease but a state of *dynamic balance* between the body, mind, spirit, and environment.

By integrating Ayurveda and Yoga into education, lifestyle, and healthcare, we are not just preserving a legacy—we are *crafting a sustainable, holistic future* for humanity.

As India takes pride in her ancient sciences and the world opens up to their wisdom, it is the responsibility of the current generation to practice, preserve, and promote them with *integrity and scientific rigor*.

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Reflective Questions:

1. How would you define the Indian Knowledge System in your own words?
2. In what ways does IKS differ from modern Western knowledge systems?
3. How can understanding IKS help in creating a more holistic view of education and life?
4. Why is it important to preserve and study ancient Indian knowledge?
5. What relevance does ancient wisdom hold in solving contemporary global issues?
6. Can ancient Indian knowledge systems coexist with modern scientific approaches?
7. What are some of the key components of the Indian Knowledge System corpus?
8. How has the transmission of IKS evolved over time?
9. Provide an introduction to the Vedic Corpus. What are the four Vedas, and what is the primary focus or content of each?
10. Outline the development and distinctive features of the Indian philosophical systems as part of IKS. Mention at least three schools of thought.
11. What challenges are involved in interpreting ancient texts in today's context?
12. How does Ayurveda differ from modern biomedicine in its understanding of health?
13. How is traditional knowledge defined within the Indian context?
14. How does traditional knowledge differ from empirical or scientific knowledge?
15. Why is traditional knowledge often undervalued in formal education?
16. What are the main characteristics that define traditional knowledge?
17. How does the scope of traditional knowledge extend beyond just rituals or folklore?
18. In what ways can traditional knowledge contribute to sustainable living and environmental conservation?
19. What distinguishes each of the four Vedas in terms of content and purpose?
20. How do the Vedas reflect the spiritual and philosophical depth of ancient Indian civilization?
21. In what ways are the teachings of the Vedas still relevant today?
22. How did Indian philosophical systems (e.g., Nyaya, Samkhya, Vedanta) develop over time?
23. What are the unique features that distinguish Indian philosophies from other global traditions?
24. How do these philosophical systems address the nature of self, reality, and liberation?
25. What makes the Bhagavad Gita a timeless philosophical and ethical text?
26. How does the Gita address the conflict between duty and personal emotion?
27. In what ways can the teachings of the Gita be applied in modern life, including leadership and personal growth?
28. Why were the Vedangas developed as auxiliary disciplines to the Vedas?
29. How do the Vedangas enhance our understanding and application of Vedic texts?
30. In what ways can the study of Vedangas contribute to modern linguistic, scientific, or philosophical inquiry?
31. Why was precise pronunciation considered crucial in the oral tradition of the Vedas?
32. How does Shiksha reflect the deep scientific understanding of sound in ancient India?
33. Can the study of phonetics in Shiksha influence modern language learning or cognitive development?
34. How does Panini's work in Vyakarana illustrate a highly developed linguistic science?
35. What does Vyakarana tell us about the importance of structure and clarity in communication?
36. How can ancient Sanskrit grammar influence computational linguistics or AI language processing?
37. What role does meter (chandas) play in preserving the oral tradition of Vedic literature?

38. How does the use of meter contribute to the aesthetic and mnemonic value of Vedic hymns?
39. Can understanding Chandas enhance appreciation for modern poetry and music?
40. How does Nirukta help uncover deeper meanings behind Vedic terms and symbols?
41. Why is etymology important when interpreting ancient texts or scriptures?
42. Can Nirukta offer a bridge between ancient and contemporary interpretations of language?
43. How was Jyotisha used to determine auspicious timings and seasonal changes?
44. What does the integration of astronomy and spirituality in Jyotisha suggest about ancient Indian worldviews?
45. In what ways can ancient Indian astronomy inform or complement modern astronomical research?
46. Why was Kalpa considered essential in the practice of Vedic rituals?
47. How does Kalpa reflect the societal values and organizational skills of ancient India?
48. Can the ritual systems described in Kalpa be interpreted as early forms of social discipline or psychological guidance?
49. What are some additional branches (e.g., Ayurveda, Arthashastra, Natya Shastra) that form part of the broader IKS?
50. How do this knowledge streams reflect the holistic and interdisciplinary nature of Indian thought?
51. What are the benefits and challenges of integrating these traditional systems into contemporary academic or policy frameworks?
52. In what ways did Vedic philosophy influence the structure of Ayurvedic theory?
53. Why is the divine origin of Ayurveda important to its traditional authority?
54. How does the concept of the human as a microcosm appear in other knowledge systems?
55. How do the five elements combine to form the three doshas?
56. In what ways do the gunas influence mental health?
57. How does your prakṛti affect your physical and emotional tendencies?
58. Why is agni considered more important than even diet in Ayurveda?
59. What role does *Pragyāparādha* play in the development of disease?
60. Why is digestion (*Agni*) so central to the Ayurvedic understanding of health?
61. How can seasonal changes contribute to disease if not properly managed?
62. Explain the concept of *Shat Kriyā Kāla* and how it helps in early disease diagnosis.
63. How does Dinacharya prevent disease even before symptoms appear?
64. What are the benefits of aligning daily routines with doshik cycles?
65. Why is waking during *Brahma Muhurta* considered ideal in Ayurveda?
66. Which elements of Dinacharya can be realistically adapted in modern urban life?
67. What role has the Indian government played in reviving Ayurveda and Yoga?
68. How has globalization influenced the practice and perception of these ancient systems?
69. What are the major challenges in integrating Ayurveda with modern healthcare?
70. In what ways can technology support the future of Ayurveda and Yoga?